City of Cambridge

COF 2026-95
FIRST IN COUNCIL
June 22, 2026

A true copy;

ATTEST:-

Paula M. Crane,
Interim City Clerk

City of Cambridge

A communication was received from Andrew Putnam, Superintendent of Urban Forestry and Landscapes, transmitting a presentation related to the Urban Forest Master Plan.

Cambridge Urban Forest Master Plan
Health and Environment Committee

5 Year Update

June 22, 2026

Cambridge Urban Forest Master Plan
Health and Environment Committee

Agenda:

- Background
- Progress towards Goals and New Targets
- Plan for the Next 5 Years
- Questions / Comments

Cambridge Urban Forest Master Plan
Health and Environment Committee

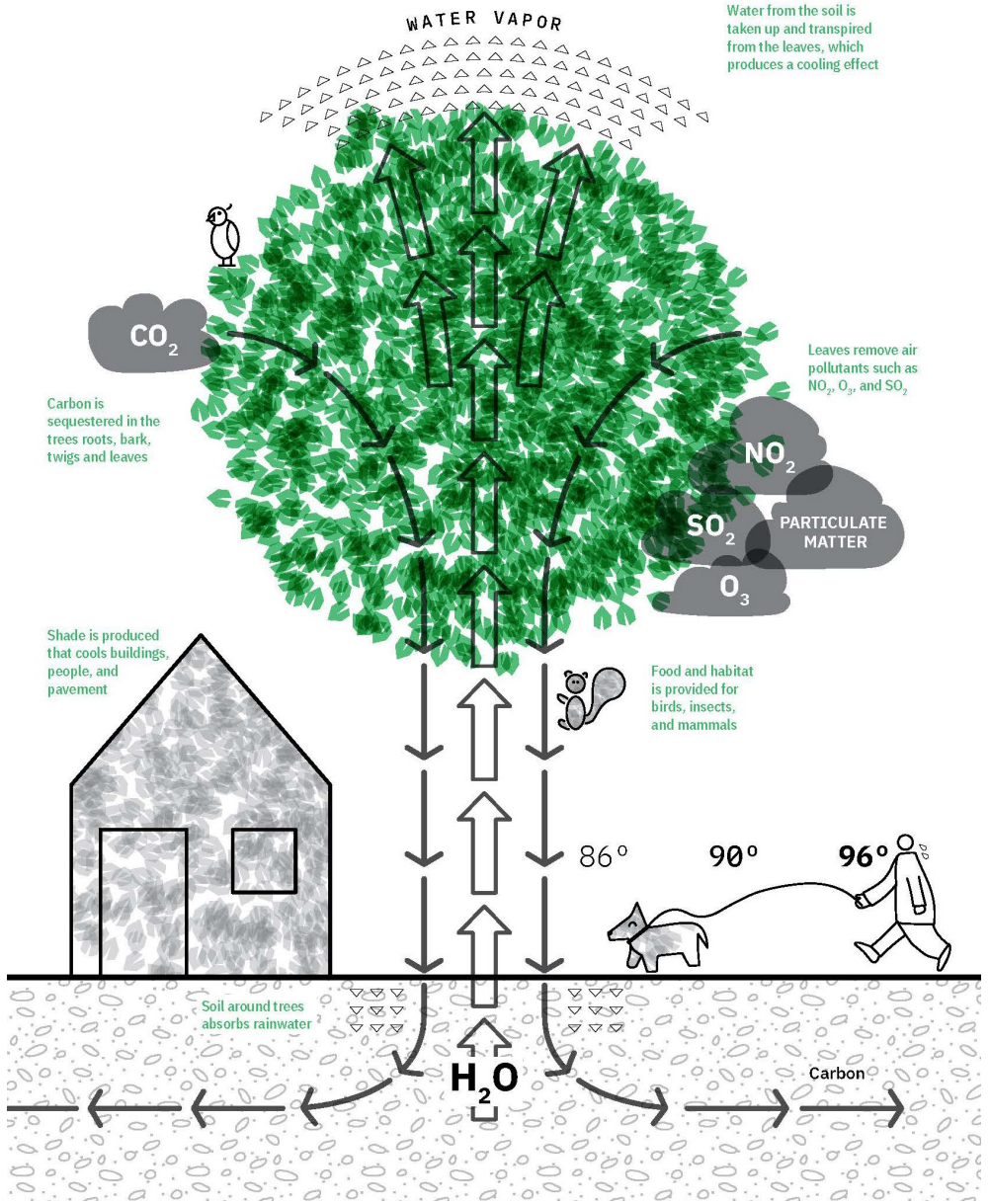
Agenda:

- Background

Cambridge Urban Forest Master Plan

A health forest is a
vital part of a healthy city

We all share responsibility
to care for our urban forest



Background and Process

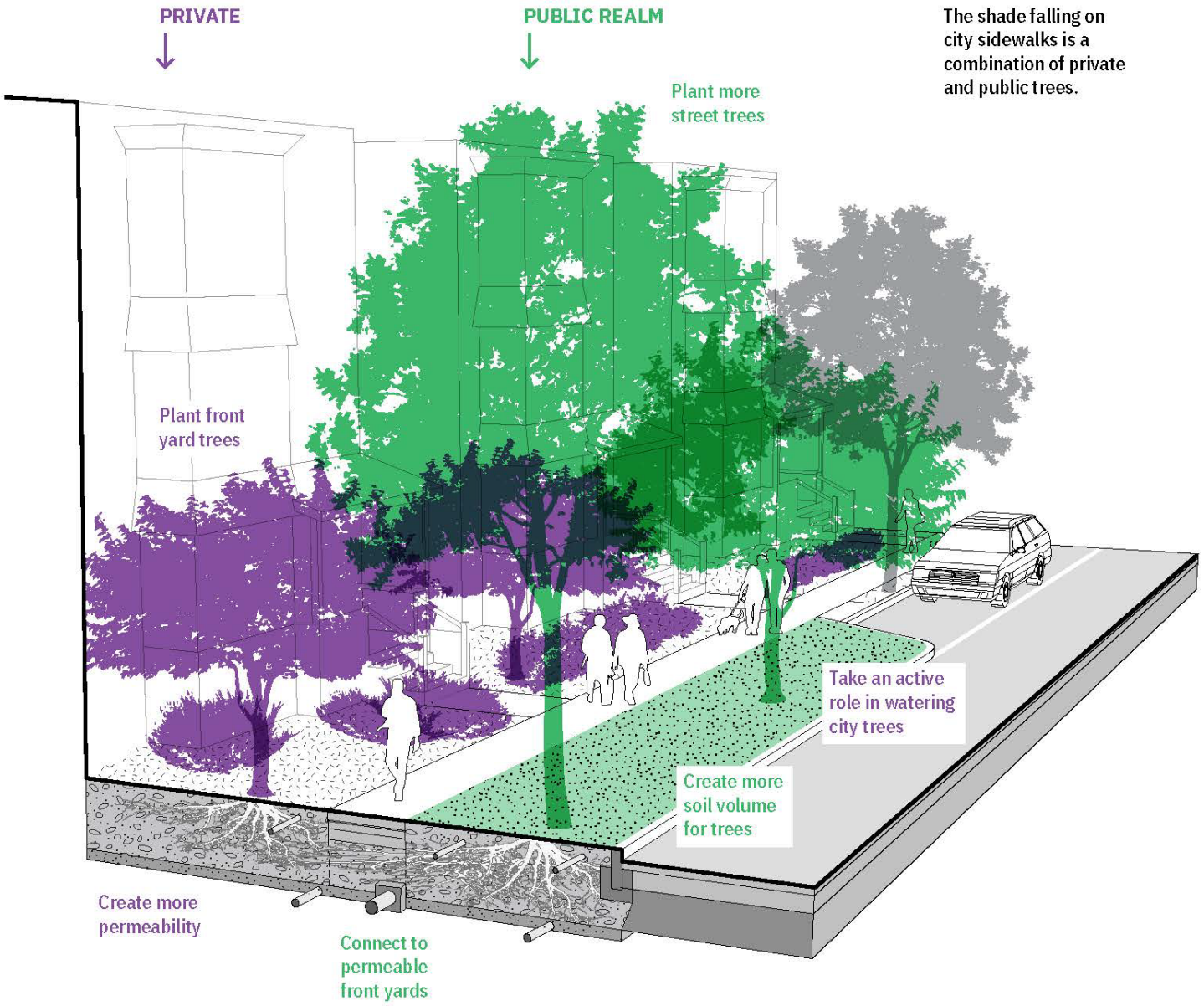
Builds on the findings of the Cambridge Climate Change Vulnerability Assessment (CCVA) and supports the goals of the Climate Change Preparedness and Resilience (CCPR) study

Supports the Core Values of Envision Cambridge

- Livability & Diversity
- Equity
- Economic Opportunity
- Sustainability & Resilience
- Community Health and Wellbeing
- Learning

Goals

Grow canopy to maintain and improve the livability, resiliency, and well-being of Cambridge and its residents.



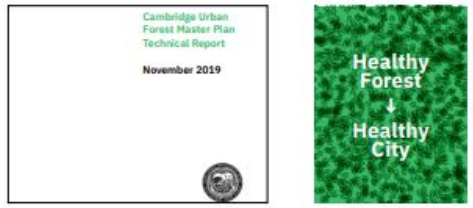
Goals of the 5- Year Update

1. **Evaluate progress since the 2019 Urban Forest Master Plan,**
   focused on the goals and priorities set forth in the UFMP

2. **Evaluate and recommend new strategies or modifications to existing strategies,**
   across policy, City practices and design, and outreach and engagement

3. **Set priorities for the next 5 years,**
   identifying the actions that will be most impactful and feasible

Timeline



2018 — 2019 — 2020 — 2021 — 2022 — 2023 — 2024 — 2025 — 2026 — 2027

CUFMP MASTER PLAN

5 YEAR UPDATE

LIDAR SURVEY

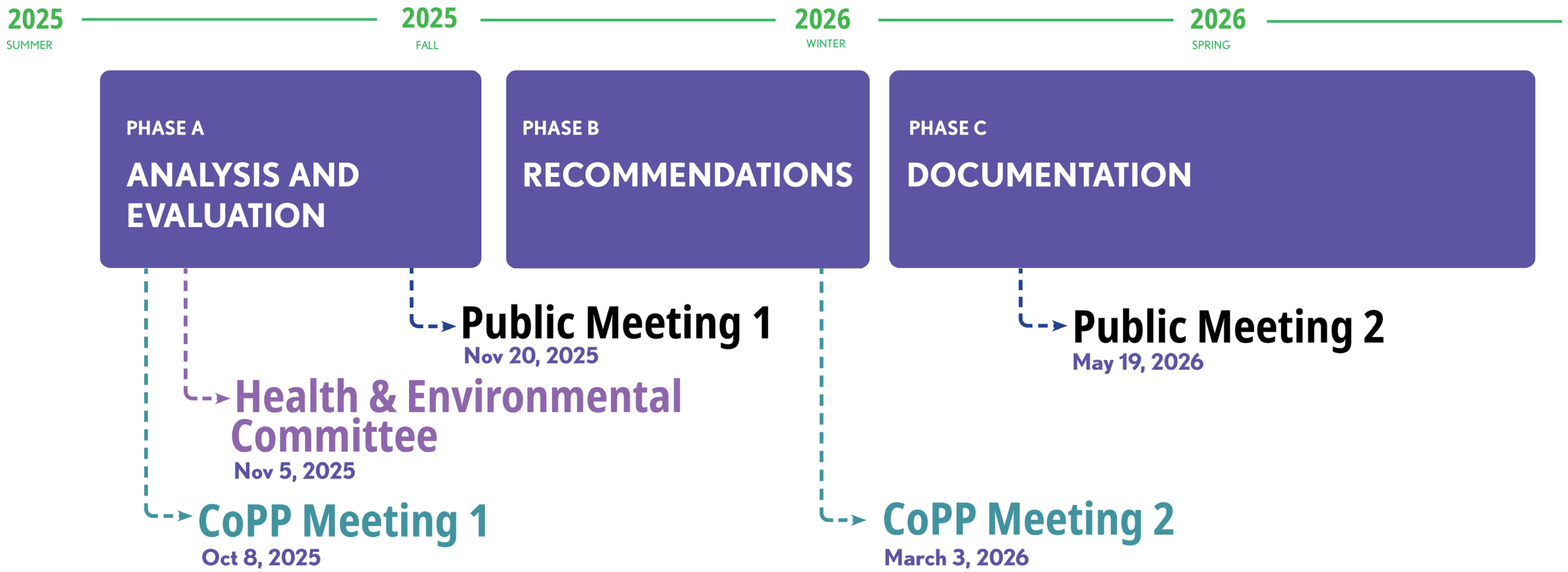
HEALTH & ENVIRONMENT COMMITTEE UPDATE

LIDAR SURVEY

JULY 15TH: CUFMP 5 YR UPDATE START

CAMBRIDGE DEPARTMENT OF PUBLIC THE WORKS

Public Meetings



Committee on Public Planting Input

Cambridge Urban Forest Master Plan
Health and Environment Committee

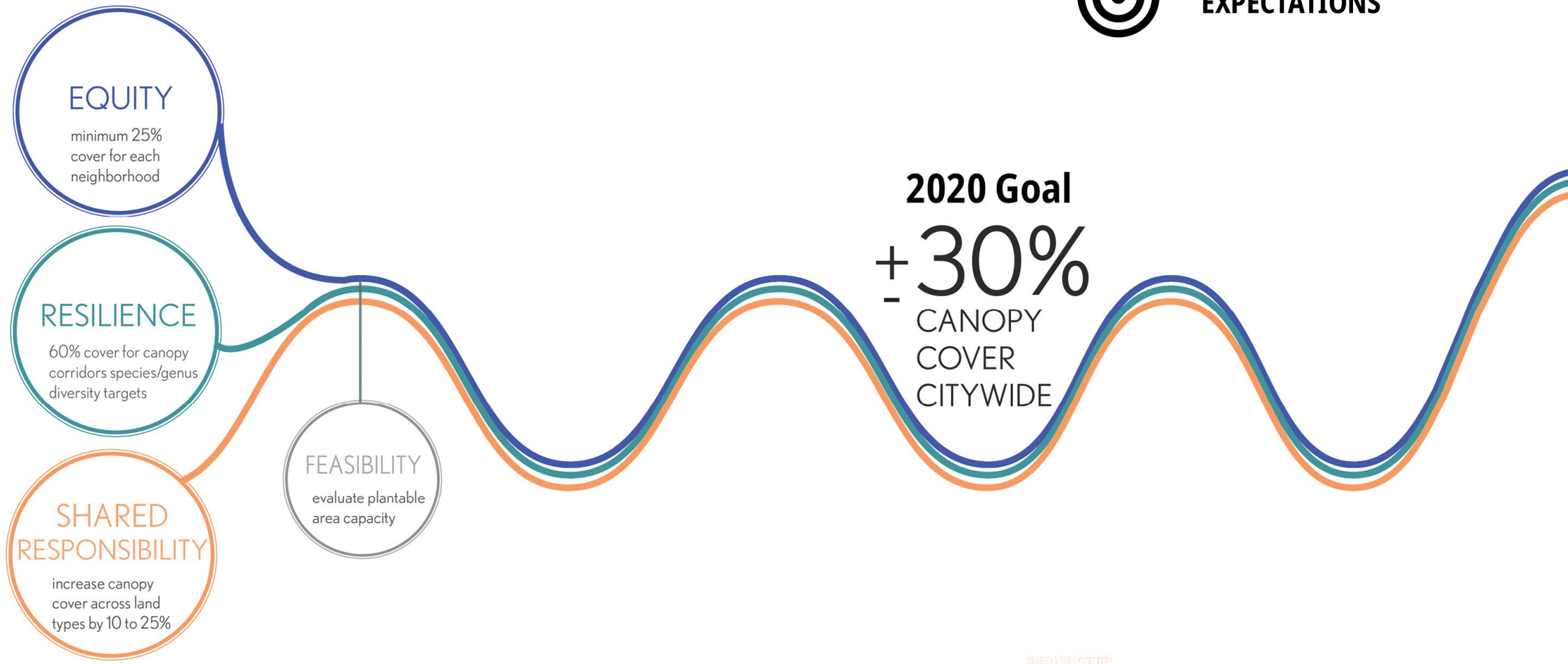
Agenda:

- Progress Towards Goals and New Targets

Measuring Progress — 5 Year Plan

Citywide Canopy Cover - 2020 Goals

MEETING/EXCEEDING EXPECTATIONS

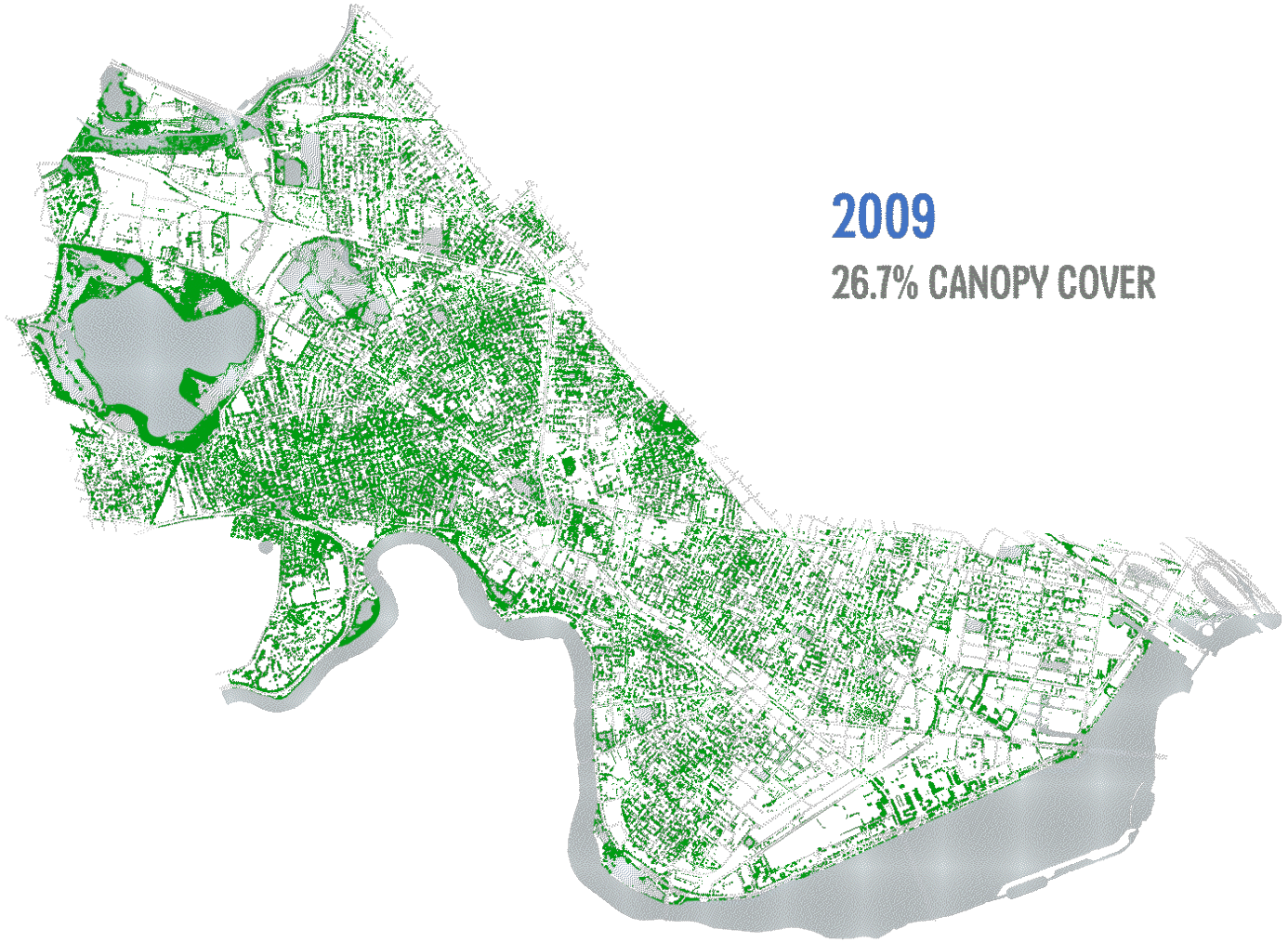


Citywide Canopy Cover - Good News

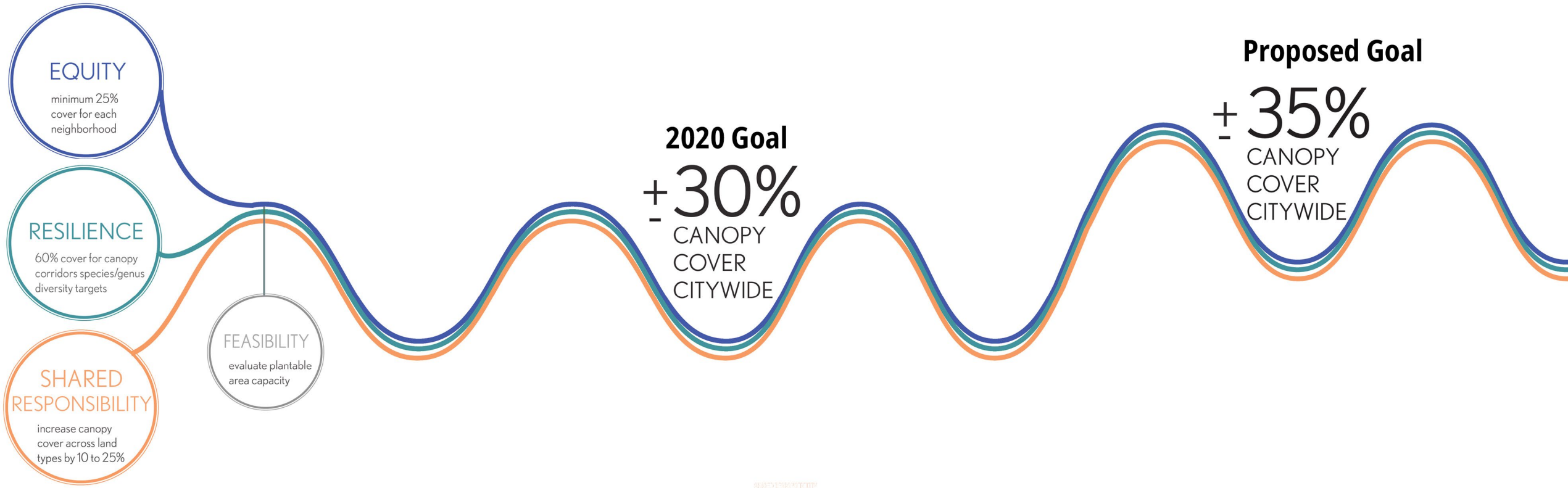
We have more canopy cover today than we did in 2009.

The rate of loss has slowed.

The rate of gain has increased.



Citywide Canopy Cover – 2026 Target



Equity - 2020 Goals

EQUITY

25% minimum canopy by neighborhood

Increase canopy cover where it is currently lacking, particularly in areas with high concentrations of minorities, the elderly, non-English speakers and those with low incomes

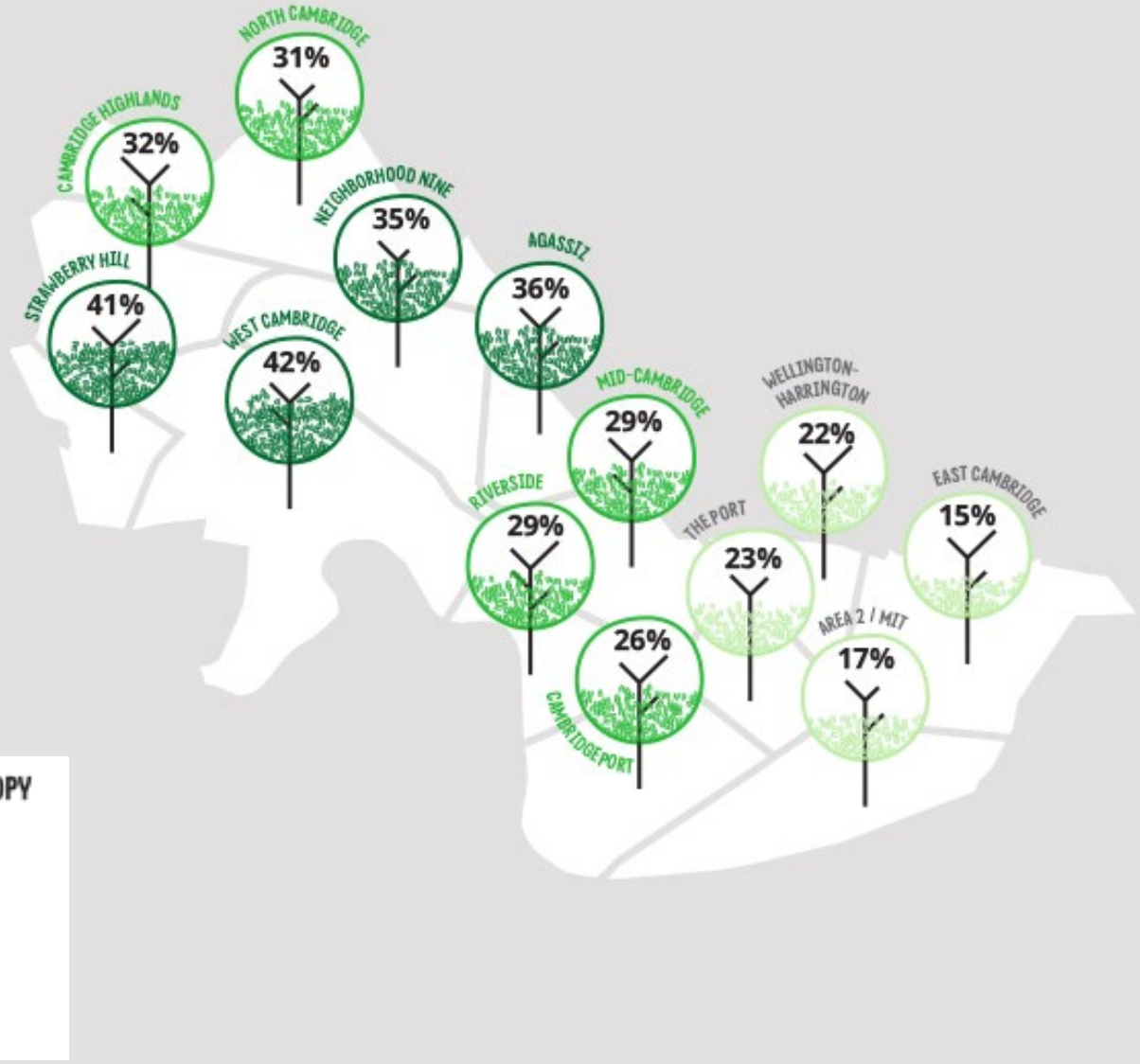
MEETING/EXCEEDING EXPECTATIONS

IMPROVING/ DEVELOPING

FALLING BEHIND

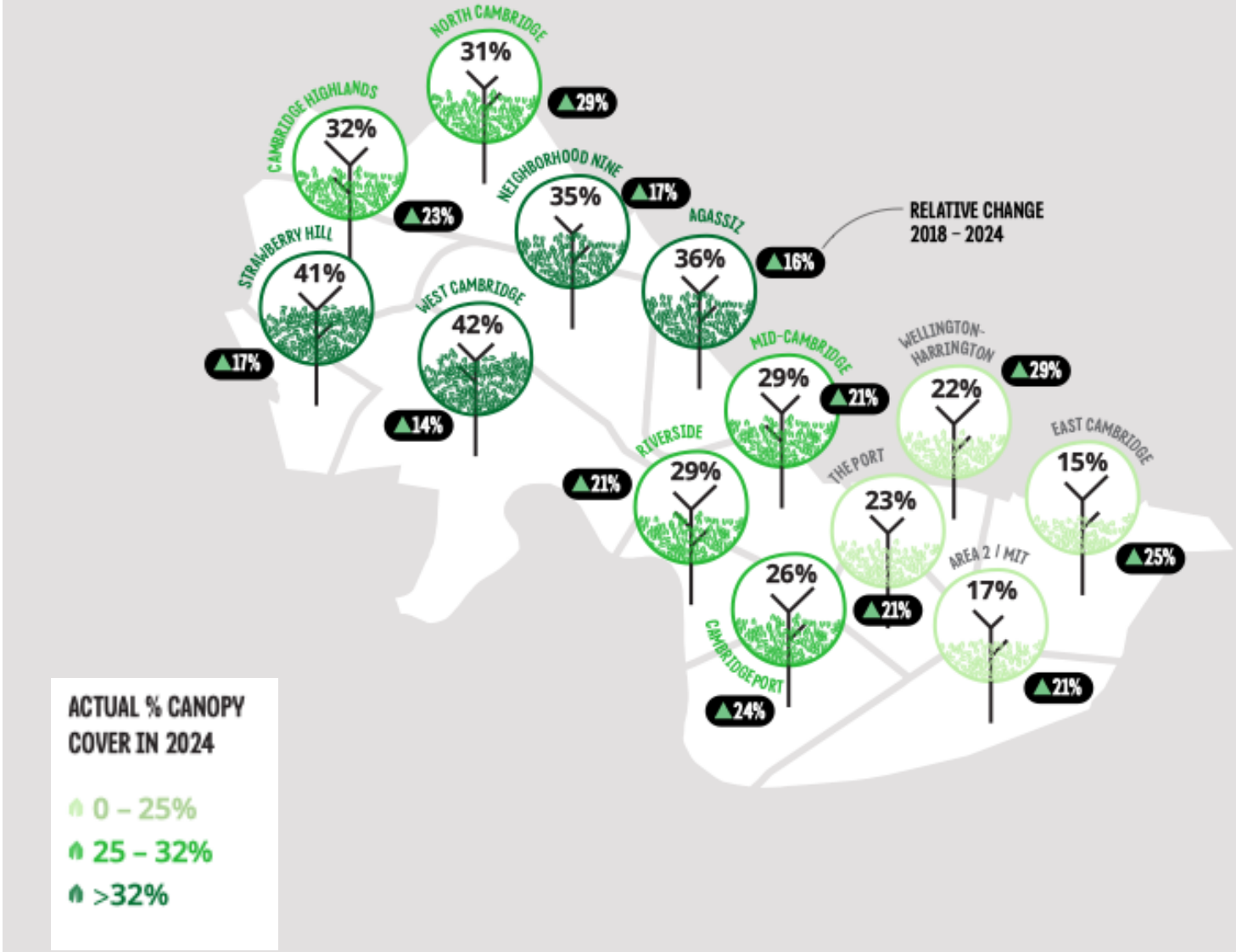
Equity - Canopy Cover

While canopy cover is still not equitably distributed …



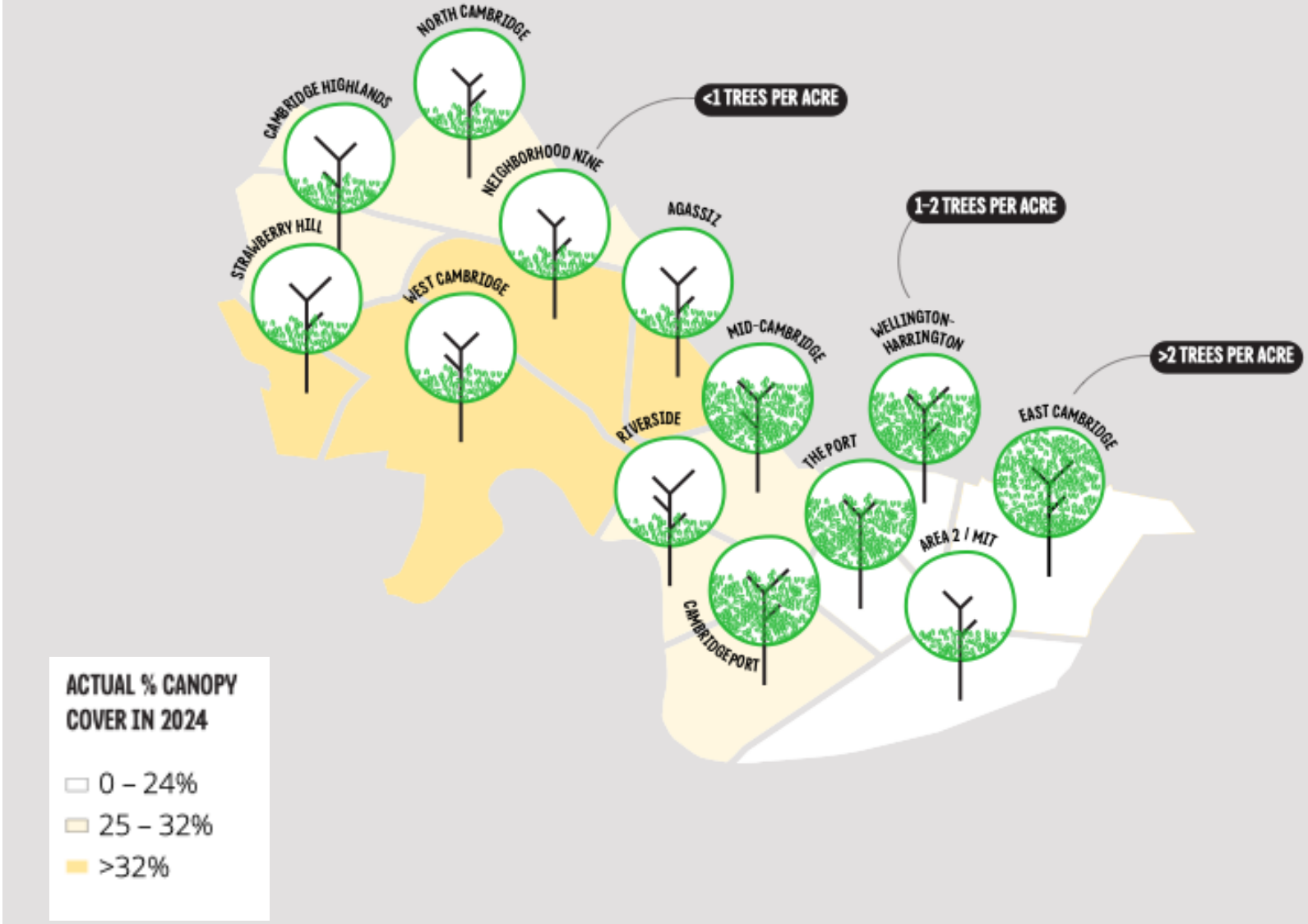
Equity - Canopy Growth

Canopy cover is growing, especially in the neighborhoods that need it the most.



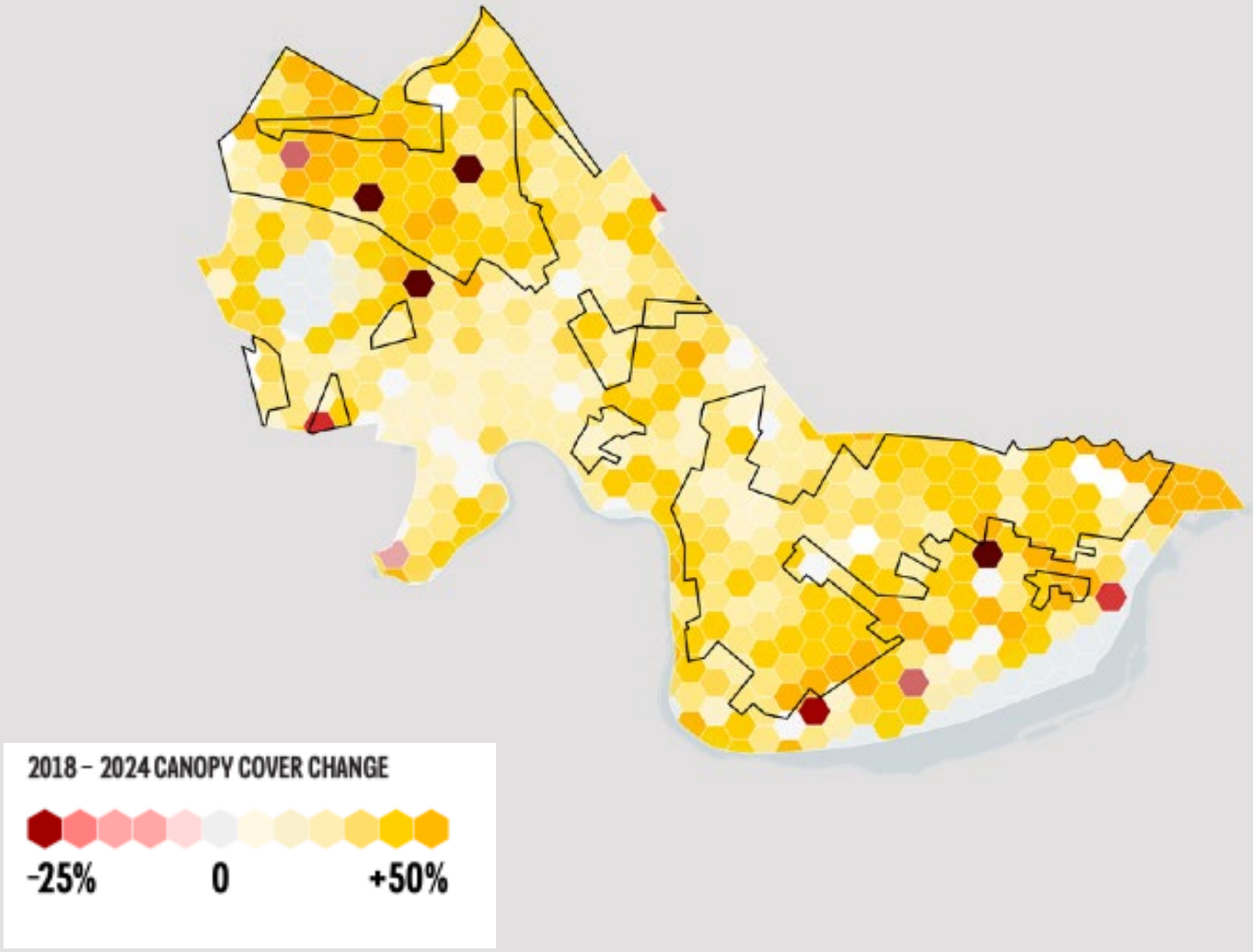
Equity - Planting Rates

Because new planting has been focused in the neighborhoods with the lowest canopy cover.



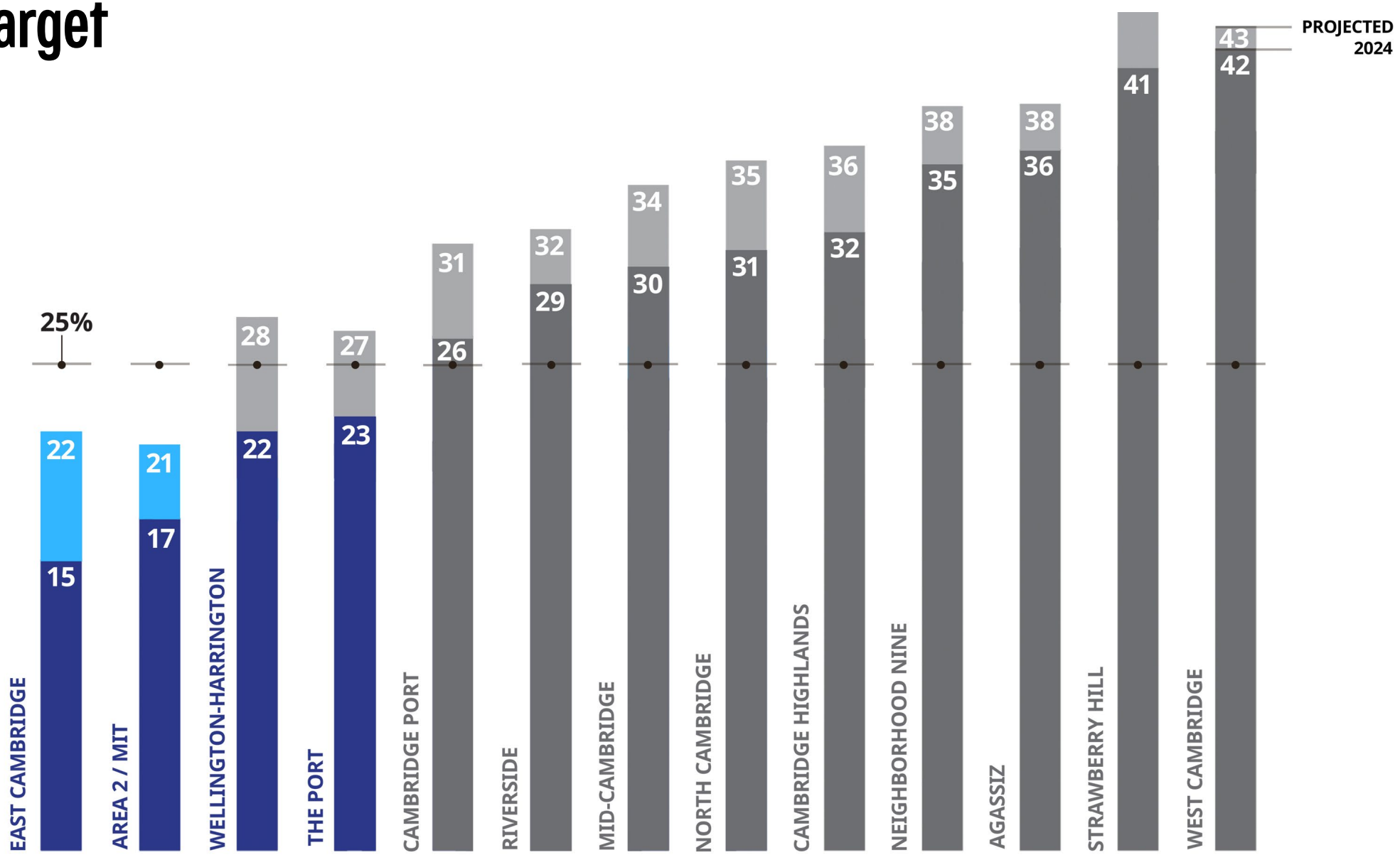
Equity - Canopy Change

The rate of growth is highest in under-canopied neighborhoods.



Equity – 2020 Target

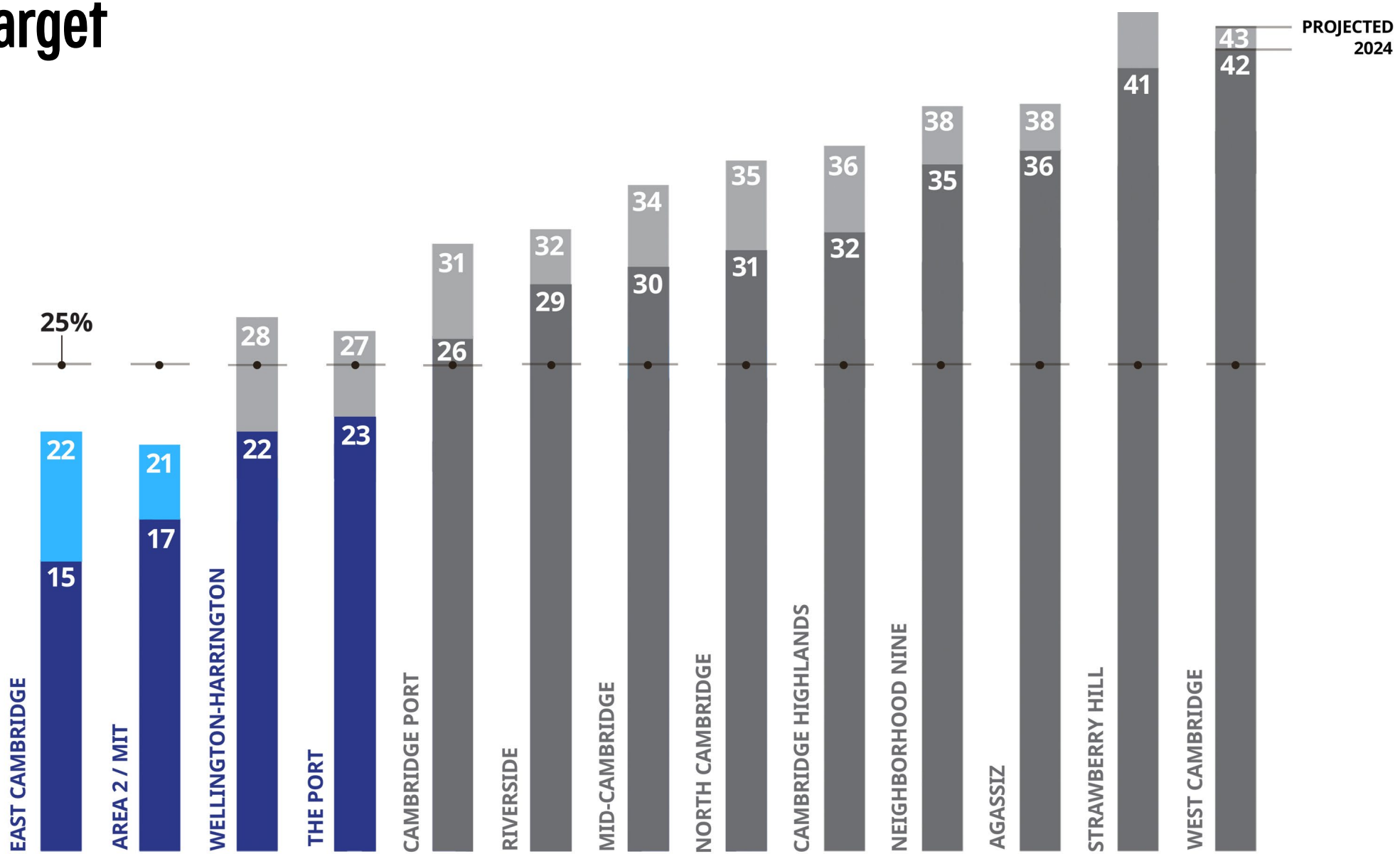
Most neighborhoods are on track to meet the 25% canopy cover target.



Equity – 2026 Target

Most neighborhoods are on track to meet the 25% canopy cover target.

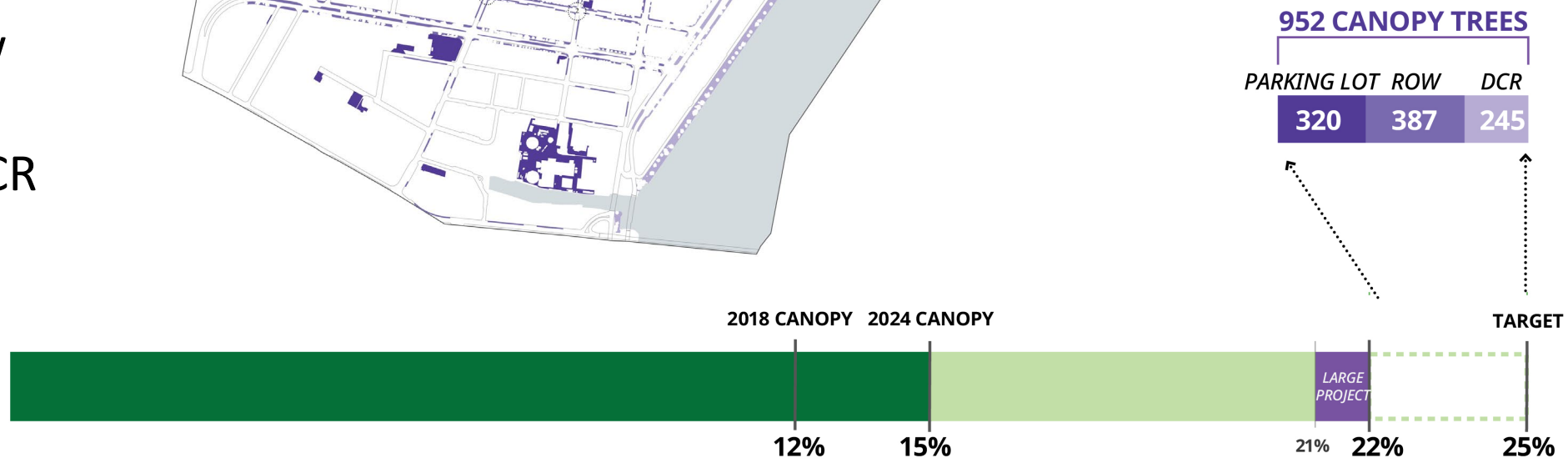
Retain 25% minimum canopy cover for each neighborhood



Equity - East Cambridge Case Study

East Cambridge has limited area available for new plantings.

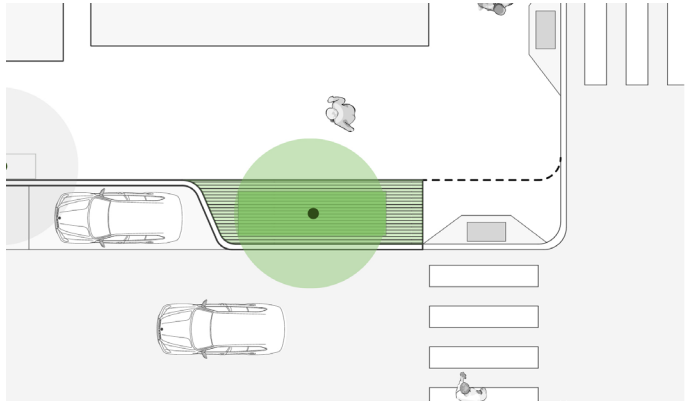
It will take a concerted effort to make space for new trees. Sidewalks, parking lots, and DCR land are key opportunity areas.



Equity - East Cambridge Case Study

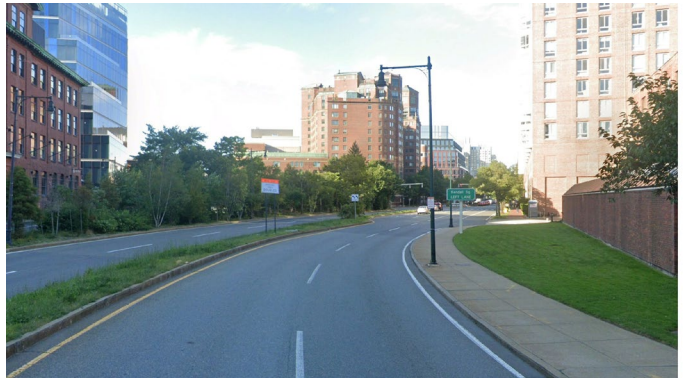
Parking Lots

Street ROW



Street Intersections

DCR Land Along Memorial Drive



DCR Land Along Edwin H. Land Blvd

Resiliency – 2020 Targets

RESILIENCY

- **60% canopy over sidewalks**
- **50% reduction of heat island hotspots**
- **Species diversification**

Enhance canopy cover in the public realm to create “cool corridors” that reduce heat island where people are impacted the most

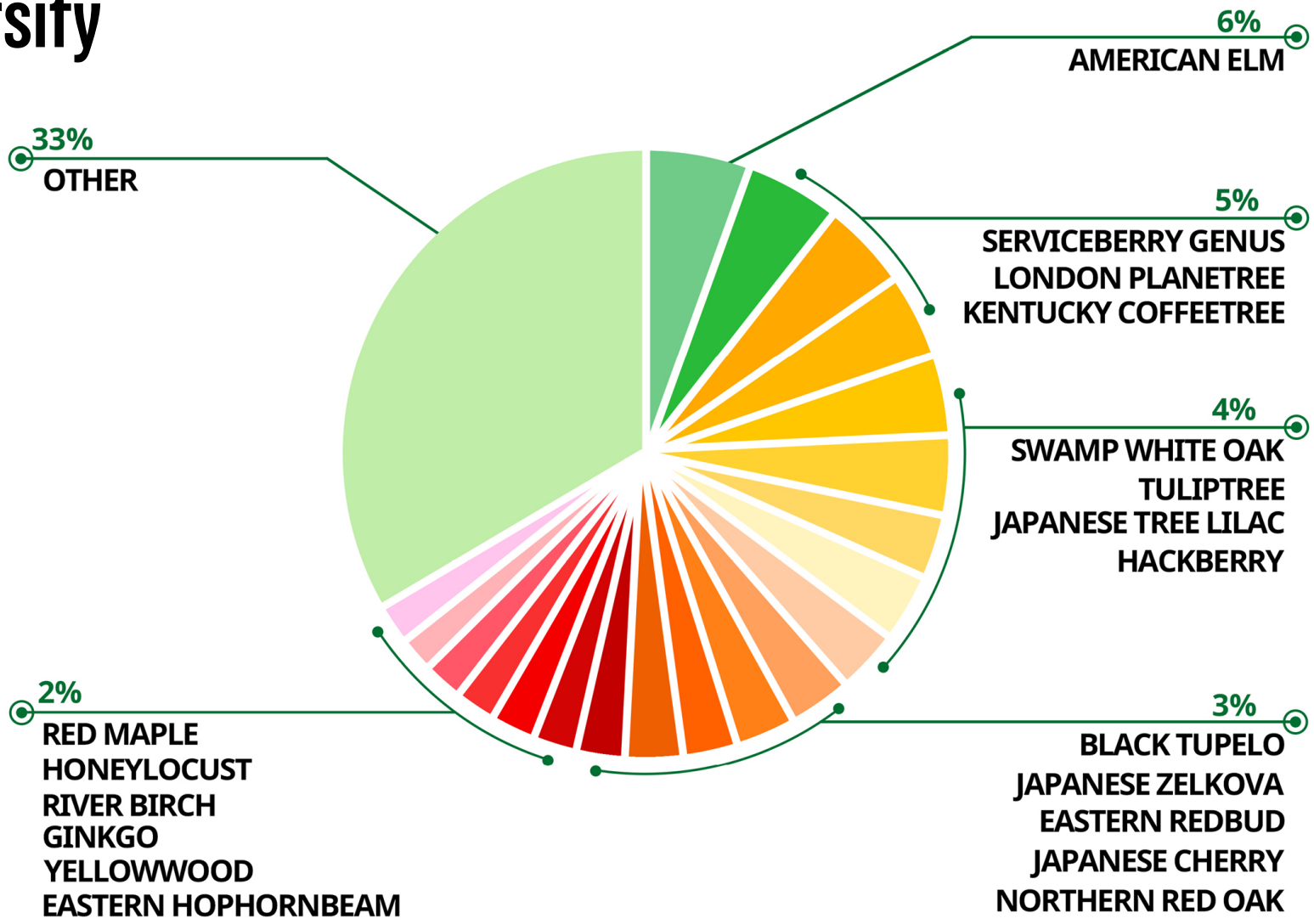
MEETING/EXCEEDING EXPECTATIONS

IMPROVING/ DEVELOPING

FALLING BEHIND

Resiliency - Canopy Diversity

The City has increased the diversity of its planting, adding 109 unique species in the last 5 years.



PLANTING BY SPECIES (2020 to 2024)

Resiliency – Species Diversity 2026 Target

No more than 10% of a single species, 20% of a genus, and 30% of a family across the city

and

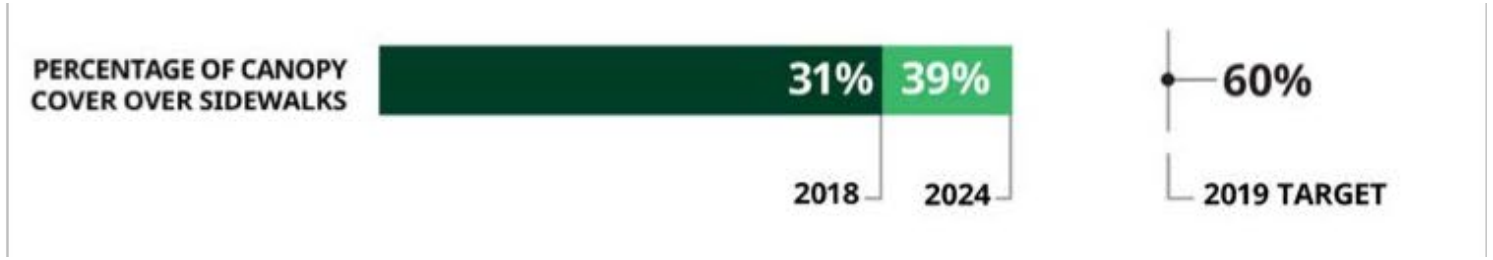
Apply for the same goal within each neighborhood

Resiliency – Genetic Diversity 2026 Target

50% of newly planted trees in parks should be native trees sourced locally (within the ecoregion)

Resiliency – Cool Corridors

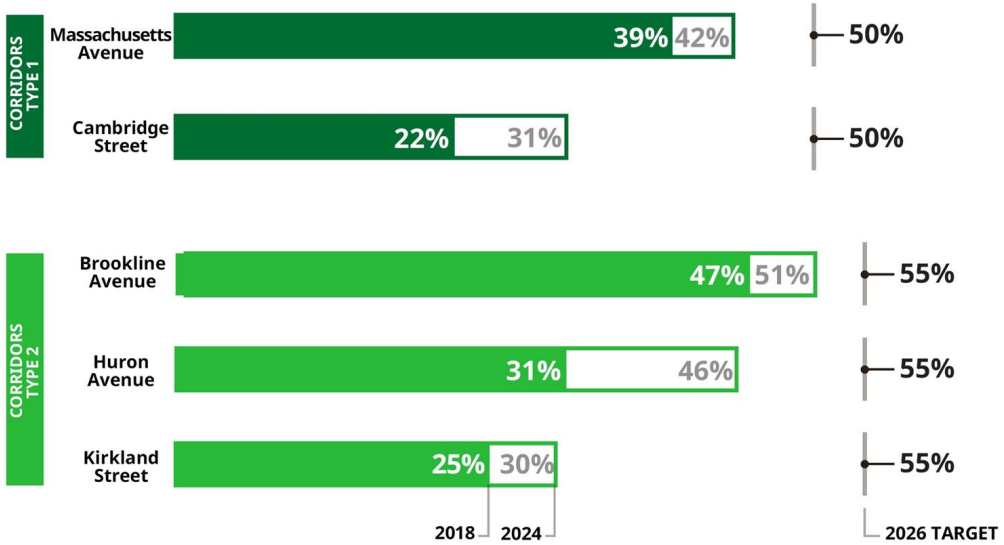
Canopy over sidewalks has grown in the last 5 years.



Resiliency - Cool Corridors 2026 Target

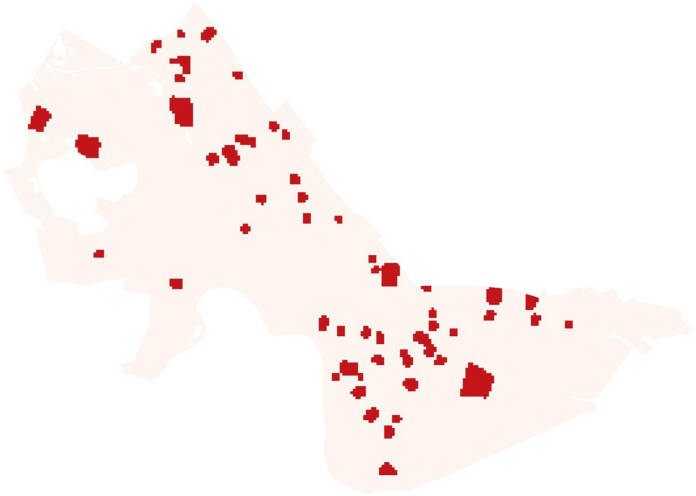
50% canopy cover over sidewalks for Type 1 corridors

55% canopy cover over sidewalks for Type 2 corridors

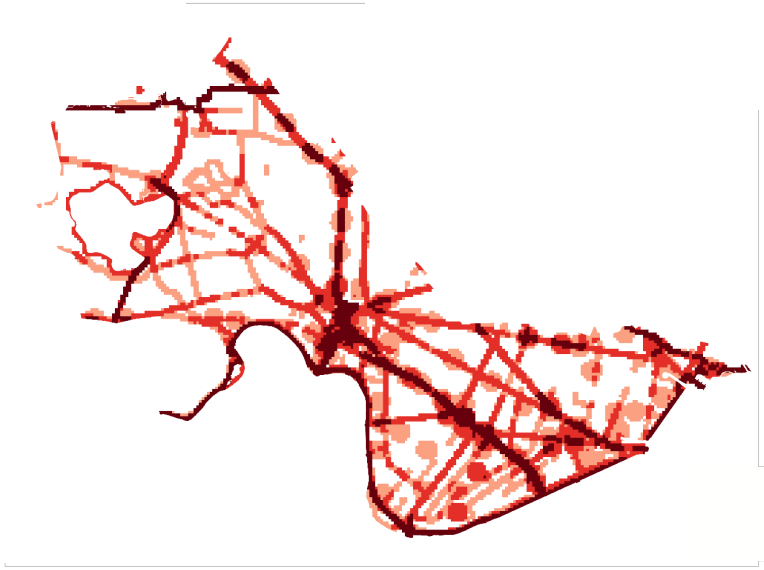


Resiliency – Priority Areas

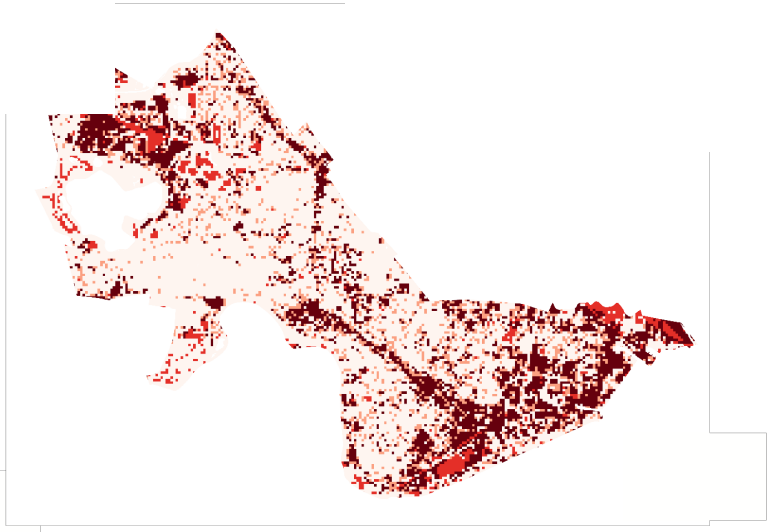
Prioritize action based on need. Areas of high need are a product of:



Populations vulnerable to heat



Commonly travelled corridors

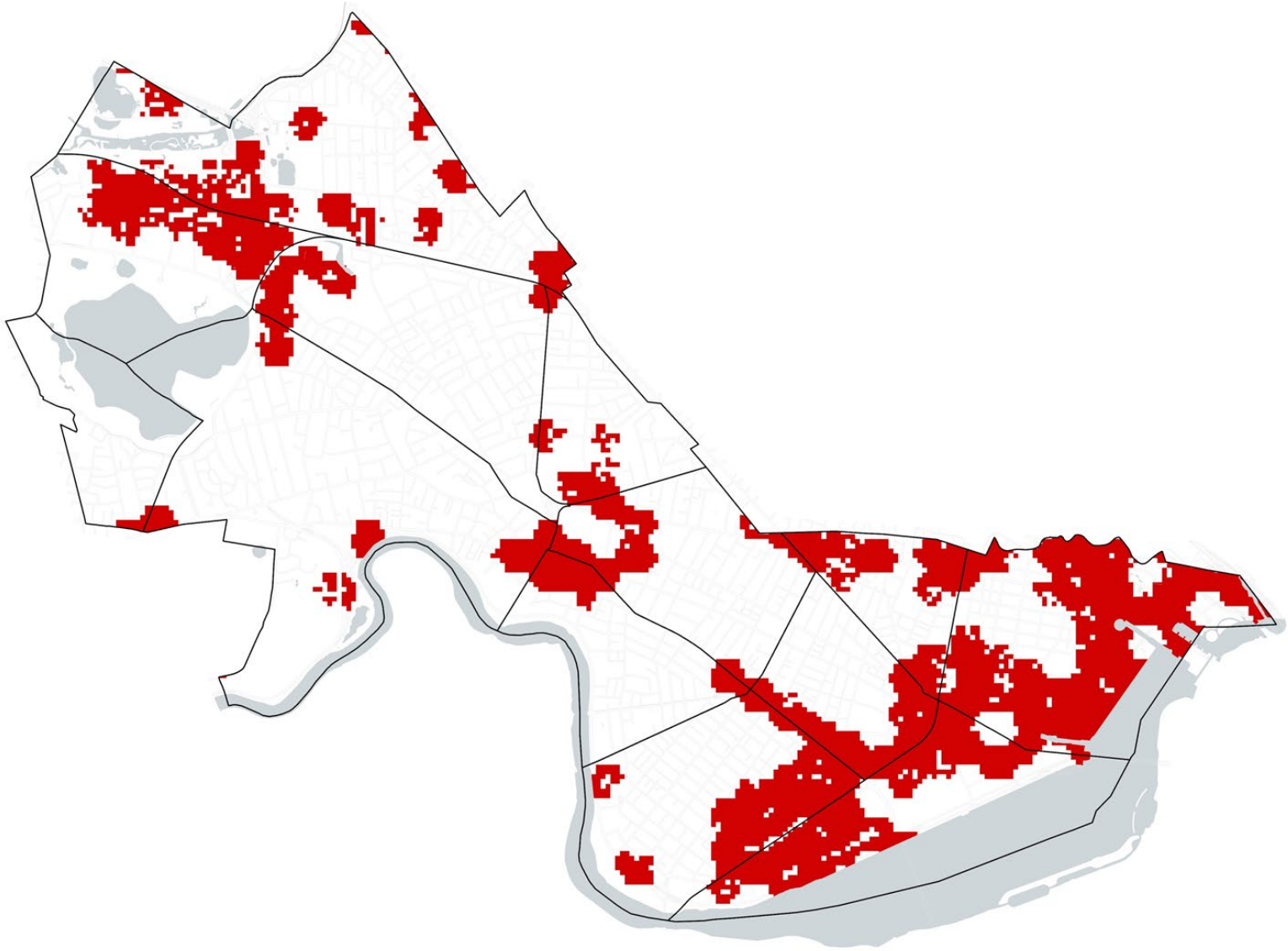


Heat islands

Resiliency – Priority Areas 2026 Target

17% canopy cover within priority areas

(currently 11%)



2020 Goals

SHARED RESPONSIBILITY

10% to 25% increase in canopy cover across all landowner types

Incentivize each landowner to increase canopy cover — a thriving forest requires mutual care by many parties

Image Credit: Billy Hickey

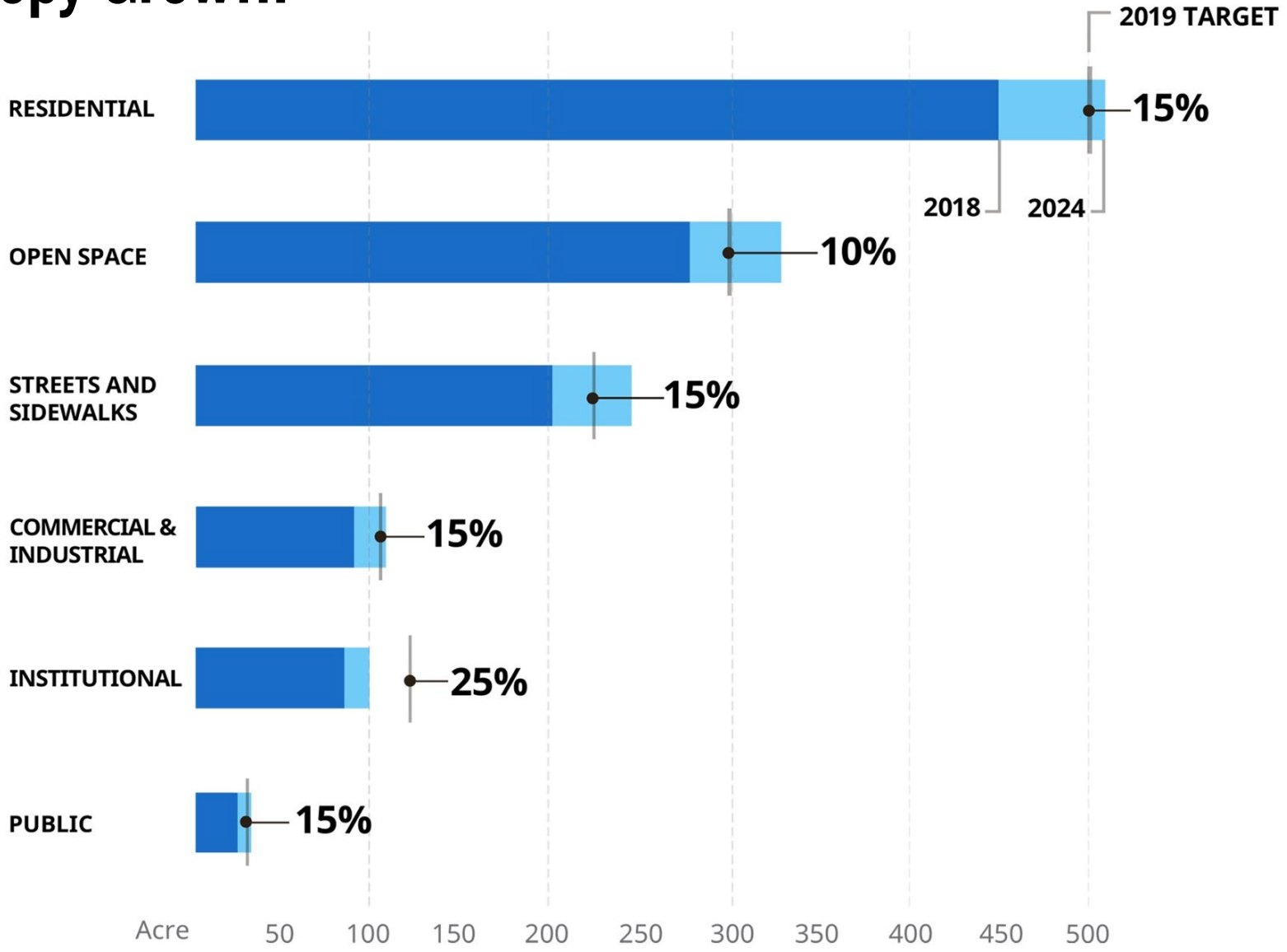
MEETING/EXCEEDING EXPECTATIONS

IMPROVING/ DEVELOPING

FALLING BEHIND

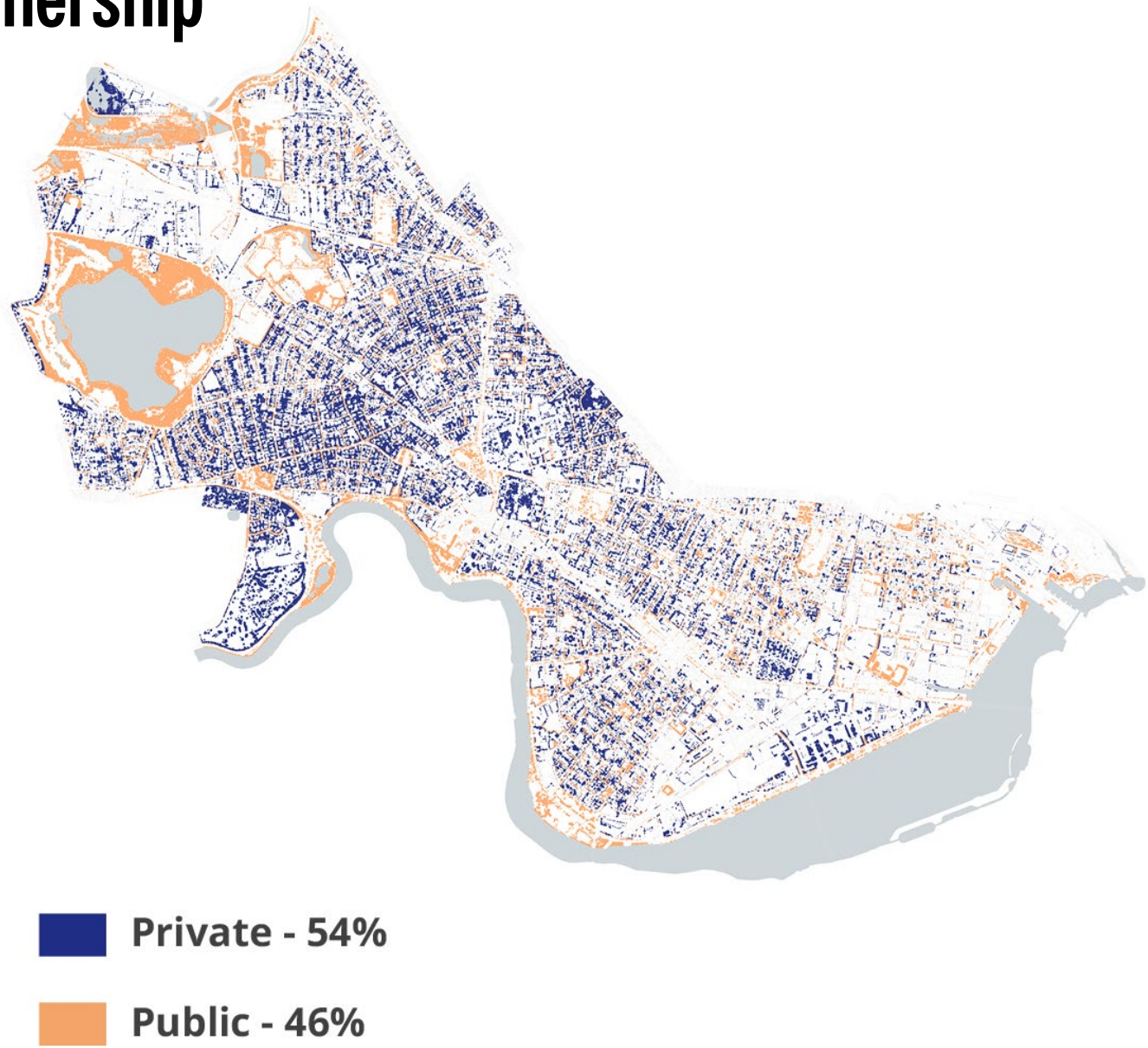
Shared Responsibility – Canopy Growth

Canopy is growing on all types of property.



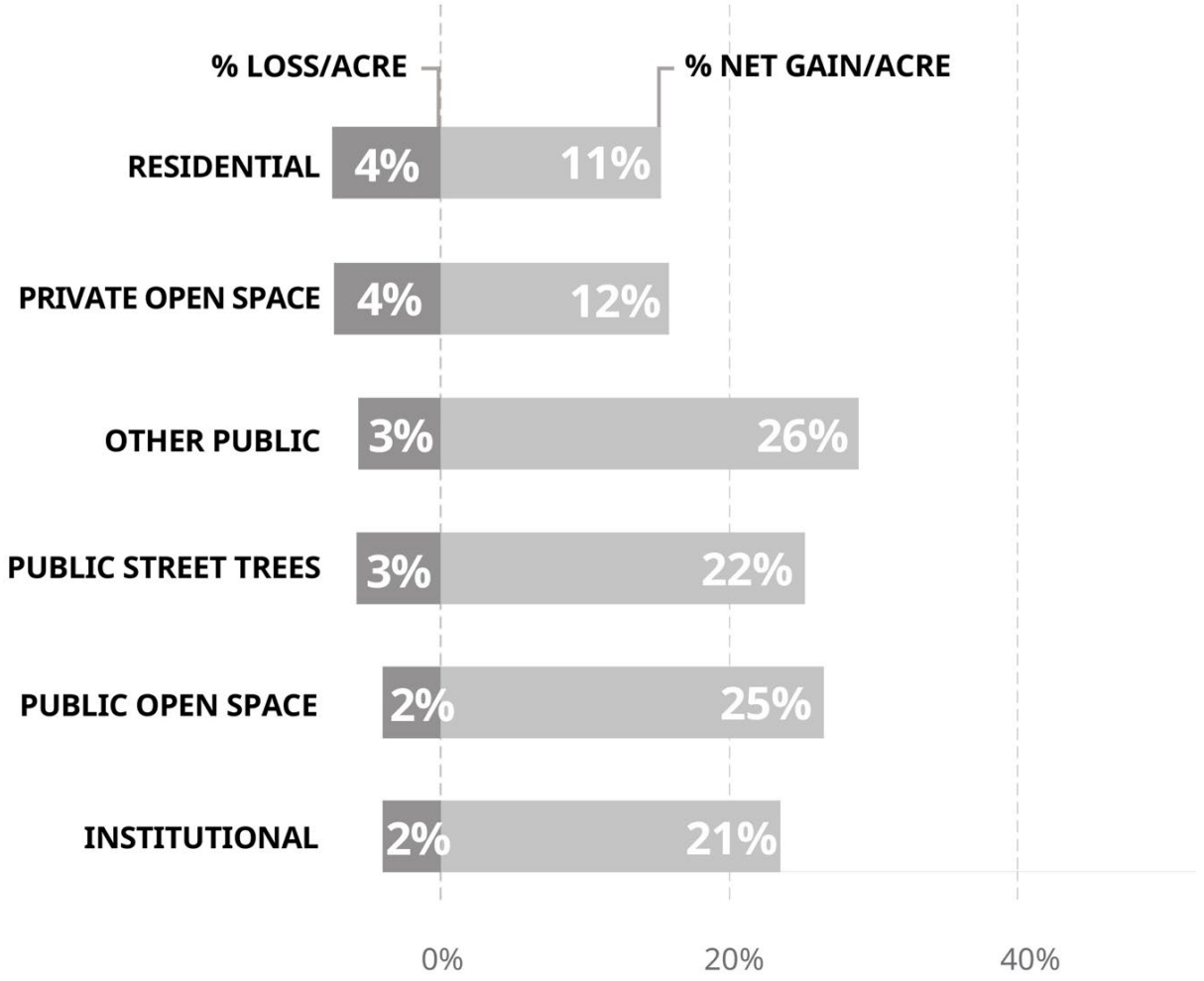
Shared Responsibility – Canopy Ownership

The majority of canopy grows on private property.



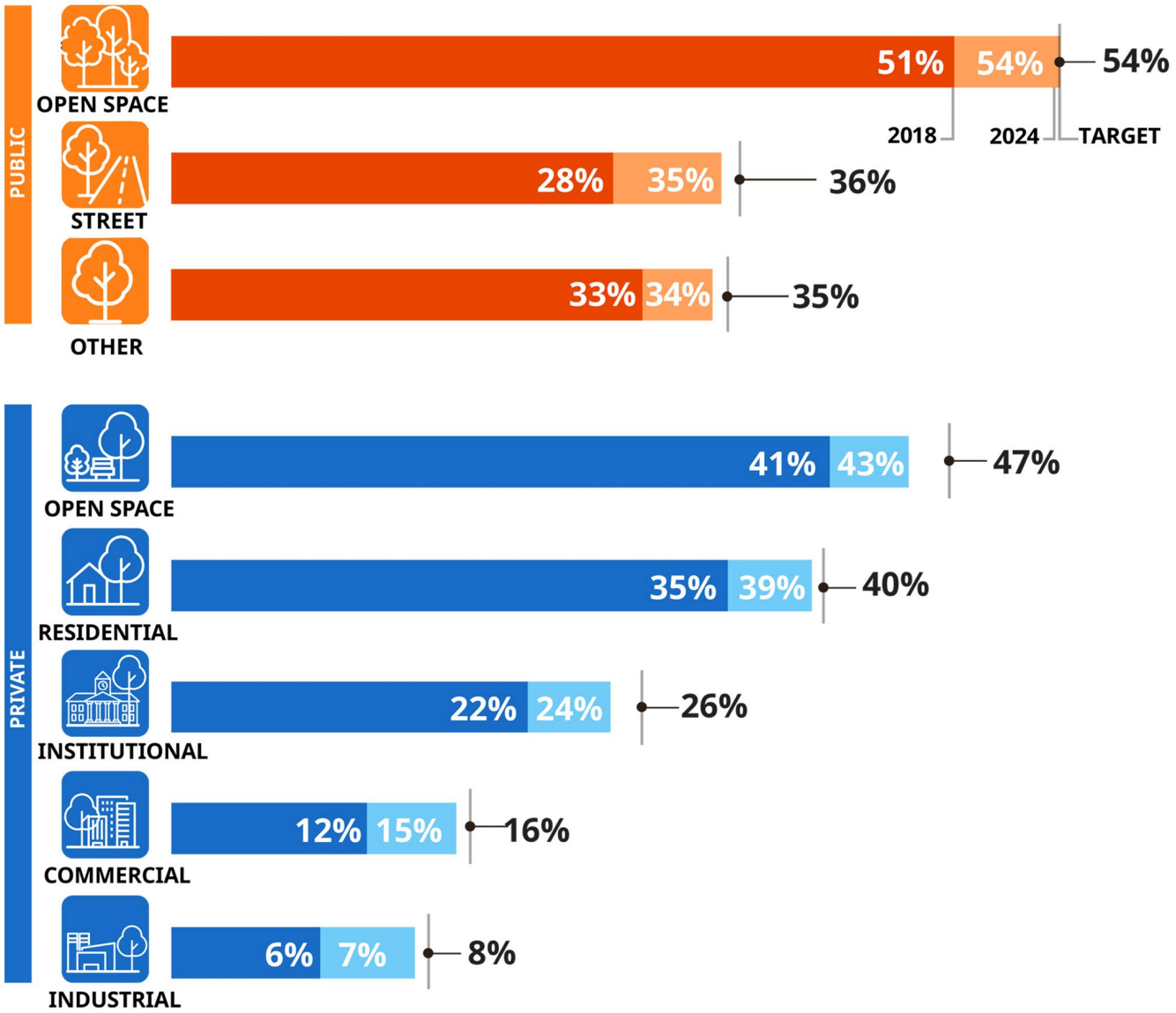
Shared Responsibility – Canopy Ownership

The most loss and least growth is occurring on residential property.



Shared Responsibility – 2026 Target

Set specific, achievable targets for public and private entities



Cambridge Urban Forest Master Plan
Health and Environment Committee

Agenda:

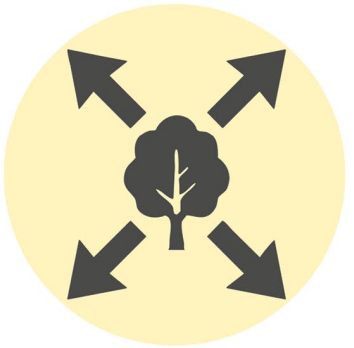
- 5 Year Plan: Key Recommendations

Key Recommendations

FOCUS ON QUALITY OVER QUANTITY

MANAGE MORE STRATEGICALLY

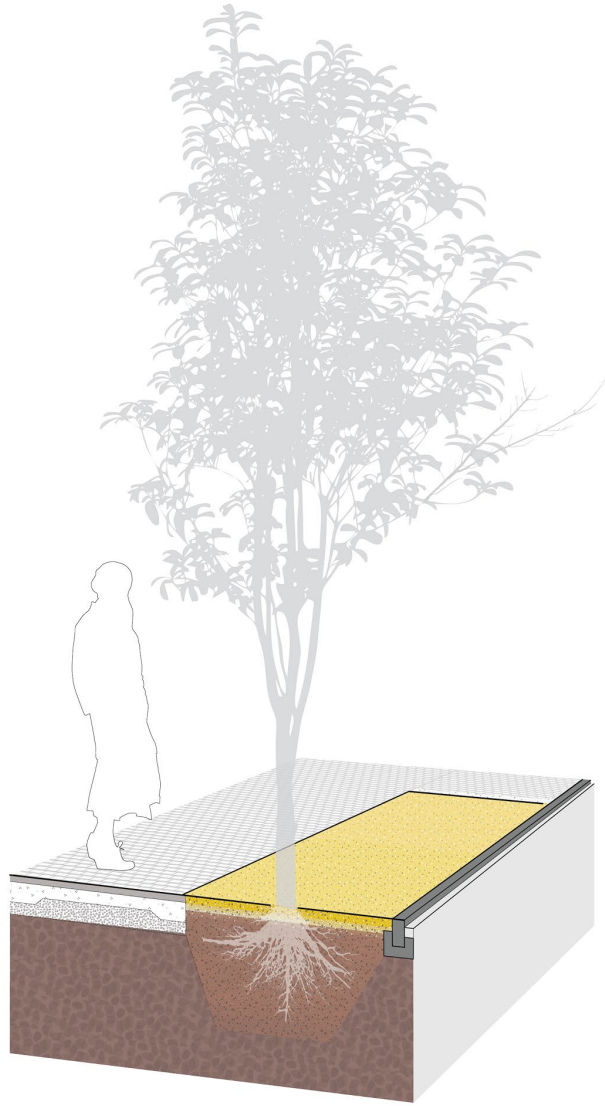
SHARE RESPONSIBILITY AND ALSO SHARE COST



BROADEN WHAT RESILIENCY MEANS

Focus on Quality over Quantity

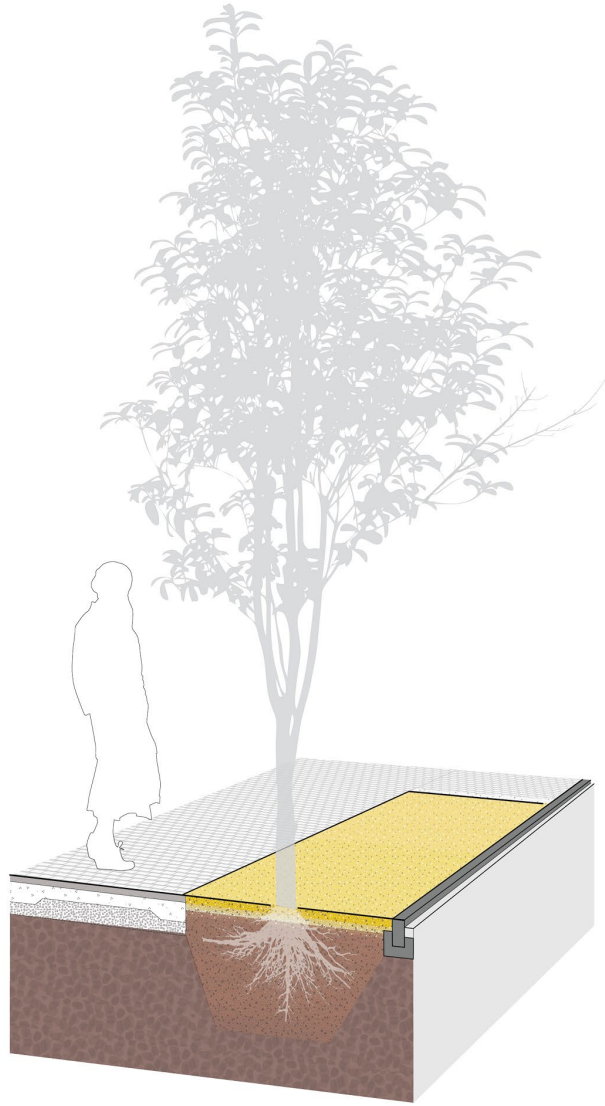
Implement enhanced planting details and care regimes, especially in the most challenging conditions, to create the best conditions for long-term growth and well-being.



Focus on Quality over Quantity

Plant trees to meet priority area, neighborhood, and land use targets, which may mean planting in more challenging places.

Reduce the 2-year mortality rate from an already-low 10% — planting 750/year instead of 1,200.



Manage More Strategically

Organize and optimize management strategies based on site type, planting condition, and risk level as opposed to neighborhood.

Share Responsibility And Also Share Cost

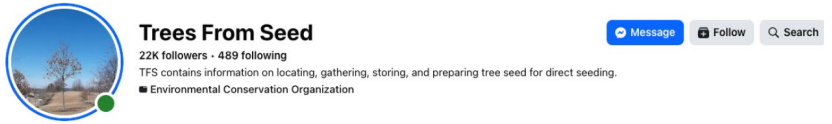
Deploy funding tools or cost sharing to incentivize new planting and retention of mature trees on private property.

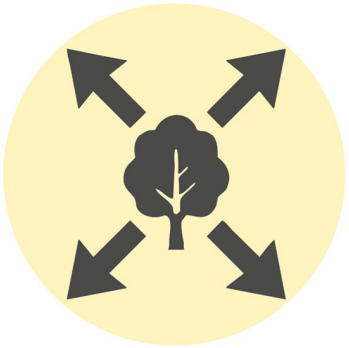
Revise policies to optimize tree protection and in-kind replacement.

Broaden What Resiliency Means

Go beyond species diversity by diversifying tree genetics.

Distribute species strategically to minimize risk from climate pressures.





Key Recommendations

Focus on Quality over Quantity

- Invest more in preparing for and planting each tree
- Optimize planting techniques for long-term success
- Institutionalize the City's tree priorities
- Formalize review practices to ensure compliance and establishment

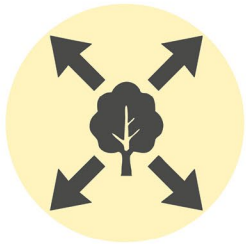
Manage More Strategically

- Expand capacity to monitor the health of the forest
- Define management zones that guide resources
- Adjust planting techniques with ongoing tree assessments

Share Responsibility and Share Cost

- Update the Tree Protection Ordinance
- Update Zoning to Incentive Planting
- Leaverage public-private partnerships to provide flexibility for planting trees on private property
- Foster stewardship through programs and partnerships
- Increase City messaging on urban forestry efforts
- Promote urban forestry education

Broaden what Resiliency Means



- Cultivate a more resilient forest
- Enable climate adaptation through genetic diversity
- Increase City-wide biodiversity
- Promote more sustainable practices

Key Recommendations

What the City can do.

- Invest more in preparing for and planting each tree
- Optimize planting techniques for long-term success
- Define management zones that guide resources
- Formalize review practices to ensure compliance and establishment
- Adjust planting techniques with ongoing tree assessments
- Increase City messaging on urban forestry efforts
- Promote urban forestry education
- Promote more sustainable practices

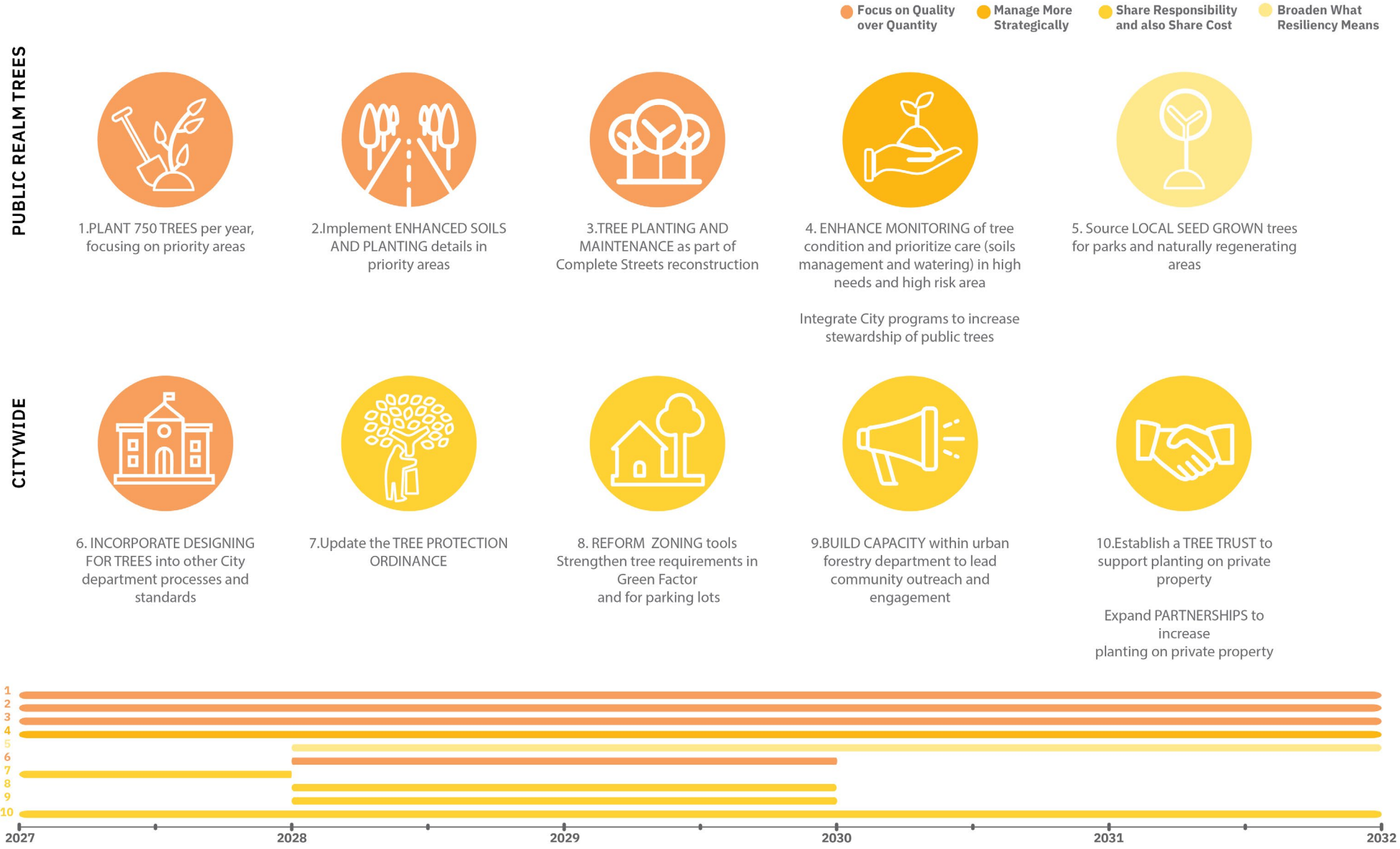
What the City can do, with residents' support.

- Expand capacity to monitor the health of the forest
- Update Zoning to Incentive Planting
- Update the Tree Protection Ordinance
- Leaverage public-private partnerships to provide flexibility for planting trees on private property
- Institutionalize the City's tree priorities

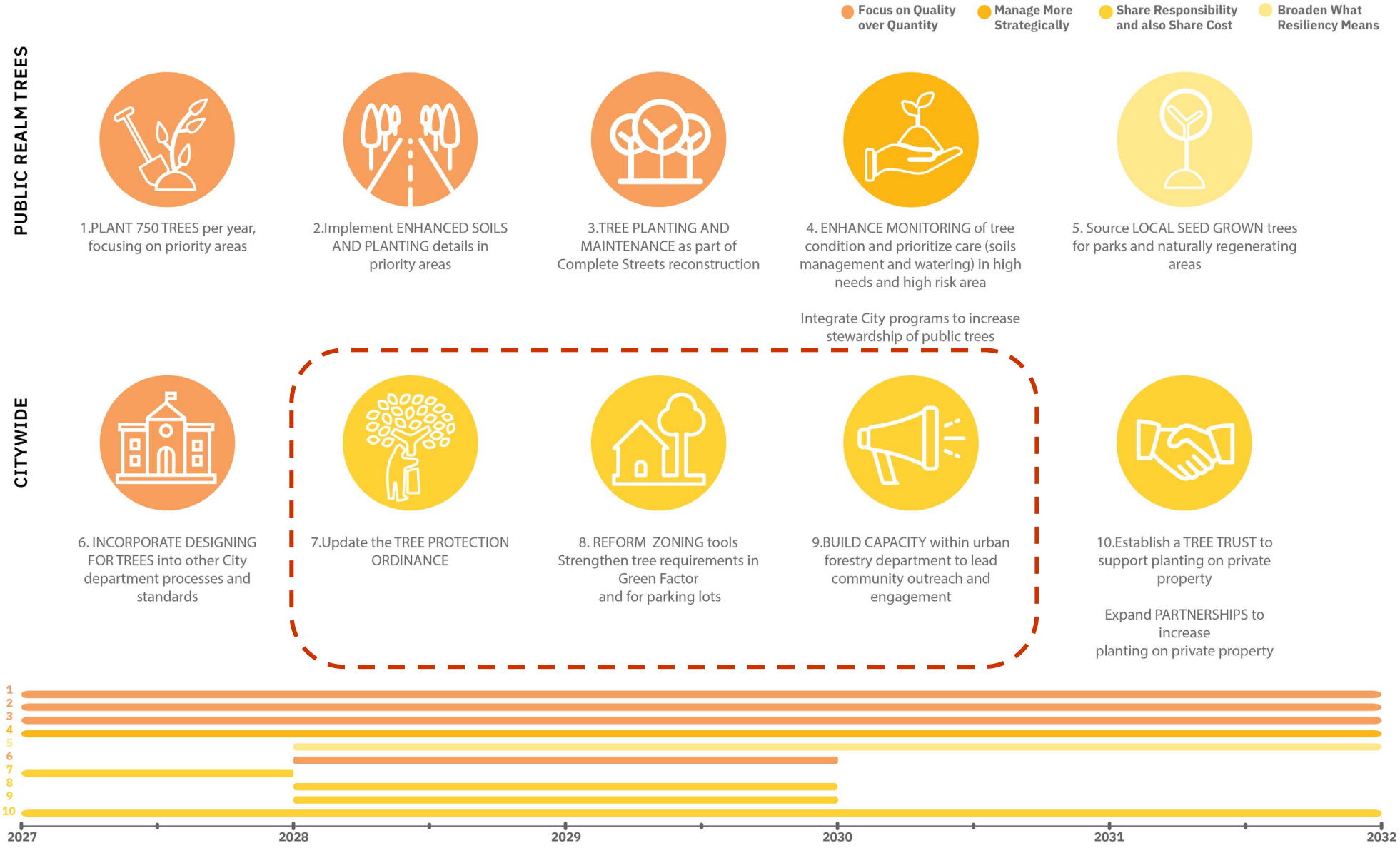
What residents can do

- Cultivate a more resilient forest
- Enable climate adaptation through genetic diversity
- Foster stewardship through programs and partnerships
- Increase City-wide biodiversity

5 Year Action Plan



5 Year Action Plan



Tree Protection Ordinance - Takeaways

Not all trees being removed under the Tree Protection Ordinance are being replanted.

Half of the canopy trees which are removed are being replaced with subcanopy trees.

TPO Recommendations: Replace more trees

75% of privately owned trees that are permitted for removal are considered "hazardous" and therefore do not require mitigation under the current ordinance.

For “hazardous” trees:

- Require 1:1 replacement so the forest continues to regenerate without being punitive by requiring caliper-inch mitigation.

- Recommend that Green Cambridge plant replacement trees where cost is an issue for the owner.

TPO Recommendations: Replace canopy with canopy

In the analysis of the replacement trees from the last five years of TPO data, 50% of canopy trees are replaced with understory trees, which provide less cooling when mature.

- Require at least one replacement tree for each removal be from the same size category and from a city-approved list of species, unless there is a significant lack of space as determined by the City Arborist.

- Add language that tree species must be climate suitable and add recommendation on genetic diversification and nursery sourcing

TPO Recommendations: Include ROW Trees Covered by Chapter 87

Protect critical root zones

- Define a tree-protection zone buffer around existing street trees, including extending onto public sidewalks.

Identify ROW trees in City applications

- Show abutting trees within 30’ of the property boundary on Building Permits and Tree Protection Plan

Protect all trees from over-pruning

- Define a threshold for maximum allowable pruning of the tree’s canopy within a specified timeframe

TPO Recommendations: Allow management of invasive trees

In naturally regenerating areas such as Fresh Pond and Alewife Reservation, trees are grown tightly together. It is onerous for the City to comply with the TPO requirements when removing large invasive trees in these areas, especially since there is limited room to replant in those areas

- For naturally regenerating areas, set a mitigation requirement of 1:1 replacement to facilitate restoration areas and removing of invasive species.

Green Factor

70 projects went through Green Factor between 2023 and 2025.

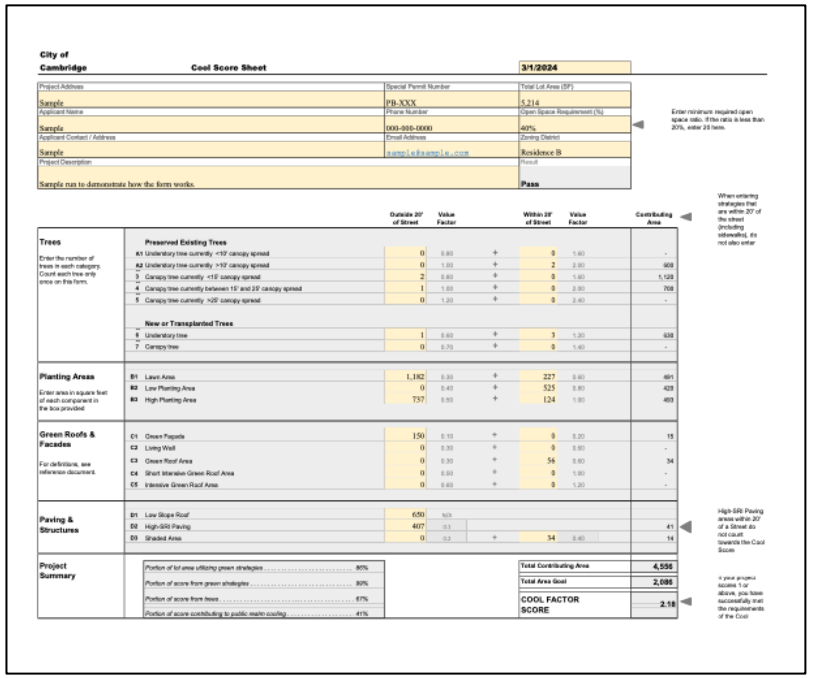
Of those projects, 858 existing trees were retained, and 1,080 new trees were planted.

At maturity, these projects will average 30% canopy cover.

18%
Existing canopy cover

→

30%
Canopy cover at maturity



City of Cambridge — Cool Score Sheet — 3/1/2024

Project Address	Special Permit Number	Total Lot Area (SF)
Sample	PB-XXX	5,214
Applicant Name	**Phone Number**	**Open Space Requirement (%)**
Sample	000-000-0000	40%
Applicant Contact / Address	**Email Address**	**Zoning District**
Sample	sample@sample.com	Residence B
Project Description		**Result**
Sample run to demonstrate how the form works.		Pass

Enter minimum required open space ratio. If the ratio is less than 20%, enter 20 here.

		Outside 20' of Street	Value Factor		Within 20' of Street	Value Factor	Contributing Area
Trees Enter the number of trees in each category. Count each tree only once on this form.	**Preserved Existing Trees**						
	A1 Understory tree currently <10' canopy spread	0	0.80	+	0	1.60	-
	A2 Understory tree currently >10' canopy spread	0	1.00	+	2	2.00	608
	3 Canopy tree currently <15' canopy spread	2	0.80	+	0	1.60	1,128
	4 Canopy tree currently between 15' and 25' canopy spread	1	1.00	+	0	2.00	708
	5 Canopy tree currently >25' canopy spread	0	1.20	+	0	2.40	-
	New or Transplanted Trees						
	6 Understory tree	1	0.40	+	3	1.20	638
	7 Canopy tree	0	0.70	+	0	1.40	-
Planting Areas Enter area in square feet of each component in the box provided	B1 Lawn Area	1,182	0.30	+	227	0.60	491
	B2 Low Planting Area	0	0.40	+	525	0.80	428
	B3 High Planting Area	737	0.50	+	124	1.00	493
Green Roofs & Facades For definitions, see reference document	C1 Green Façade	150	0.10	+	0	0.20	15
	C2 Living Wall	0	0.30	+	0	0.60	-
	C3 Green Roof Area	0	0.30	+	56	0.60	34
	C4 Short Intensive Green Roof Area	0	0.60	+	0	1.00	-
	C5 Intensive Green Roof Area	0	0.80	+	0	1.20	-
Paving & Structures	D1 Low Slope Roof	650	N/A				
	D2 High-SRI Paving	407	0.1				41
	D3 Shaded Area	0	0.2	+	34	0.40	14

When entering strategies that are within 20' of the street (including sidewalks), do not also enter

High-SRI Paving areas within 20' of a Street do not count towards the Cool Score

Project Summary		
Portion of lot area utilizing green strategies	80%	
Portion of score from green strategies	90%	
Portion of score from trees	67%	
Portion of score contributing to public realm cooling	41%	
Total Contributing Area		4,556
Total Area Goal		2,086
COOL FACTOR SCORE		2.18

If your project scores 1 or above, you have successfully met the requirements of the Cool

Green Factor Recommendations

- Broaden the applicability of the Green Factor requirements.
- Review the weighting structure of different strategies (preserving existing canopy, planting new trees) to ensure that the structure is incentivizing tree preservation and planting.
- Require / Verify compliance with minimum soil volumes for new trees.
- Track and analyze the impact of the recent zoning change that converted all residential to C-1 zoning classification.
- Require during the certificate of occupancy approvals that the design is reviewed for compliance in terms of the number and types of trees planted.

Responding to Multi-Family Residential Zoning

What can be done to protect trees?

- Strengthen the protection of trees (and critical root zones) on abutting properties
- Calibrate Green Factor to better protect existing trees
- Re-evaluate set-backs and open space definitions (green roofs and balconies)

Outreach and Engagement

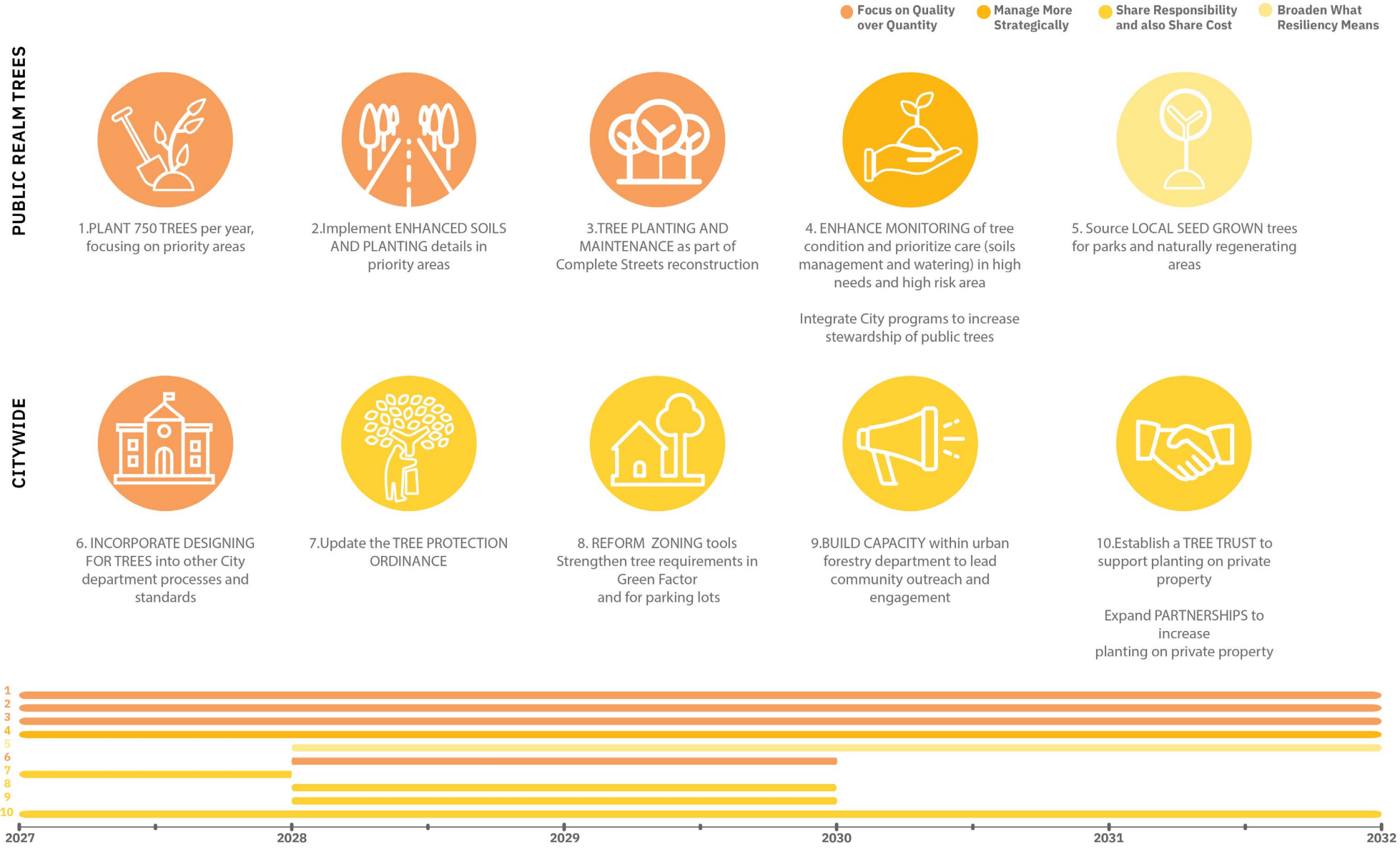
Be a Leader in Stewardship

- Join Forest Friends and care for the City's trees
- Create and Lead Neighborhood Teams

Be a Tree Advocate

- Speak up for trees at public meetings
- Voice support for policy changes recommended in this plan (TPO, GF, Tree Trust, parking lots)
- Share messages with your neighbors about the importance of sourcing locally and planting diverse species

5 Year Action Plan



Cambridge Urban Forest Master Plan
Health and Environment Committee

5 Year Update

June 22, 2026

Questions & Comments

Respond to Multi-Family Residential Zoning Change

Residential A
50% open space requirement

Residential B
40% open space requirement

→

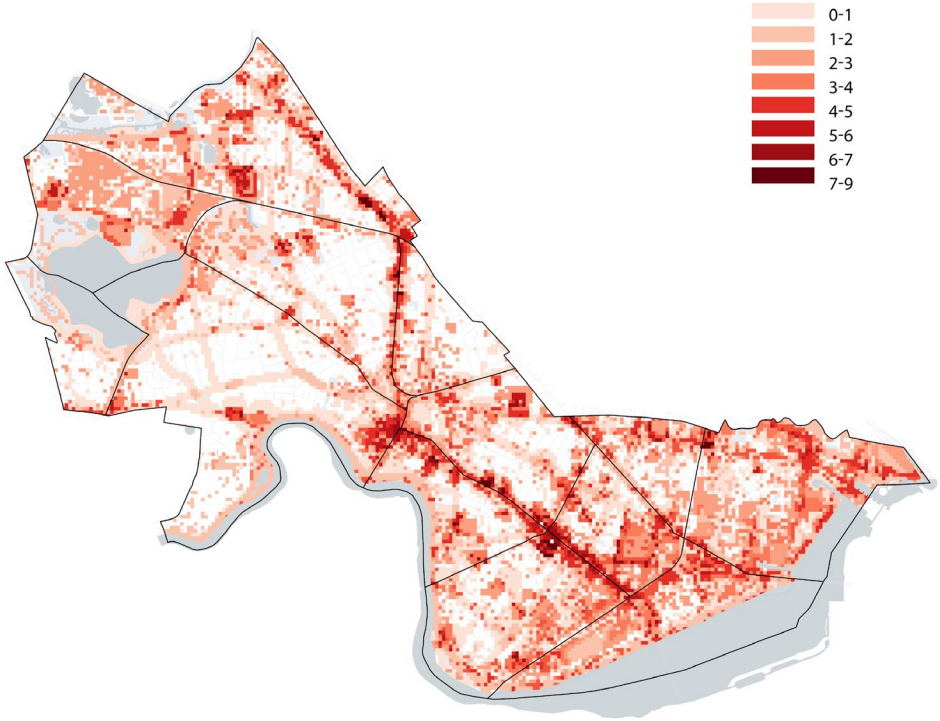
Residential C1
30% open space requirement

- If those parcels were built out (not assuming they use green roofs as open space) up to 135 acres of canopy cover could be removed, resulting in 26% canopy cover for residential areas.

- Citywide, current canopy cover is 30%. This would reduce to 28%.

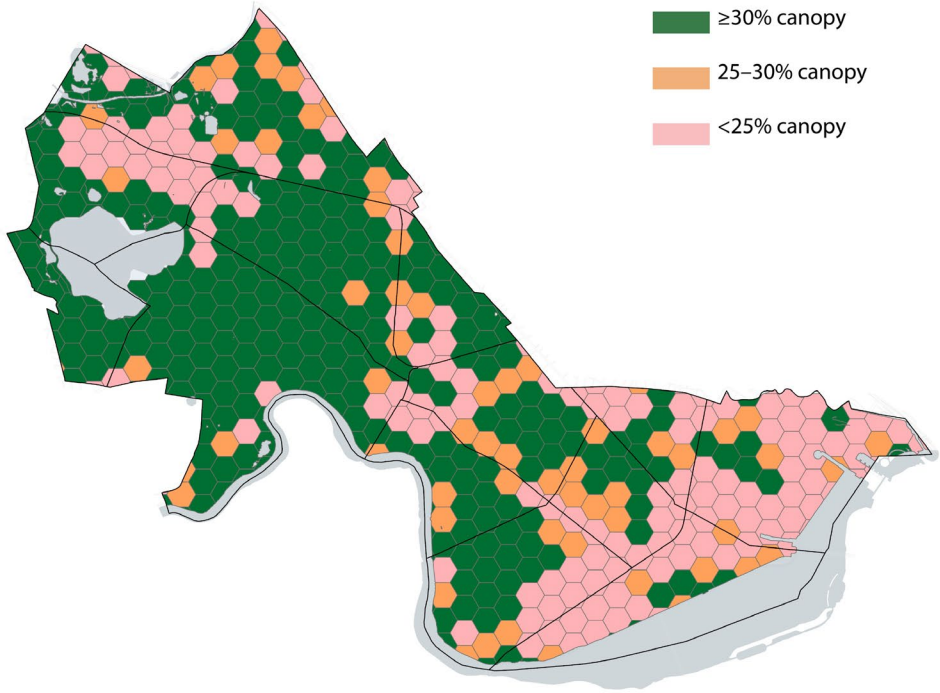
Resiliency – Priority Areas

Priority Areas are determined by:



Areas of high need

+

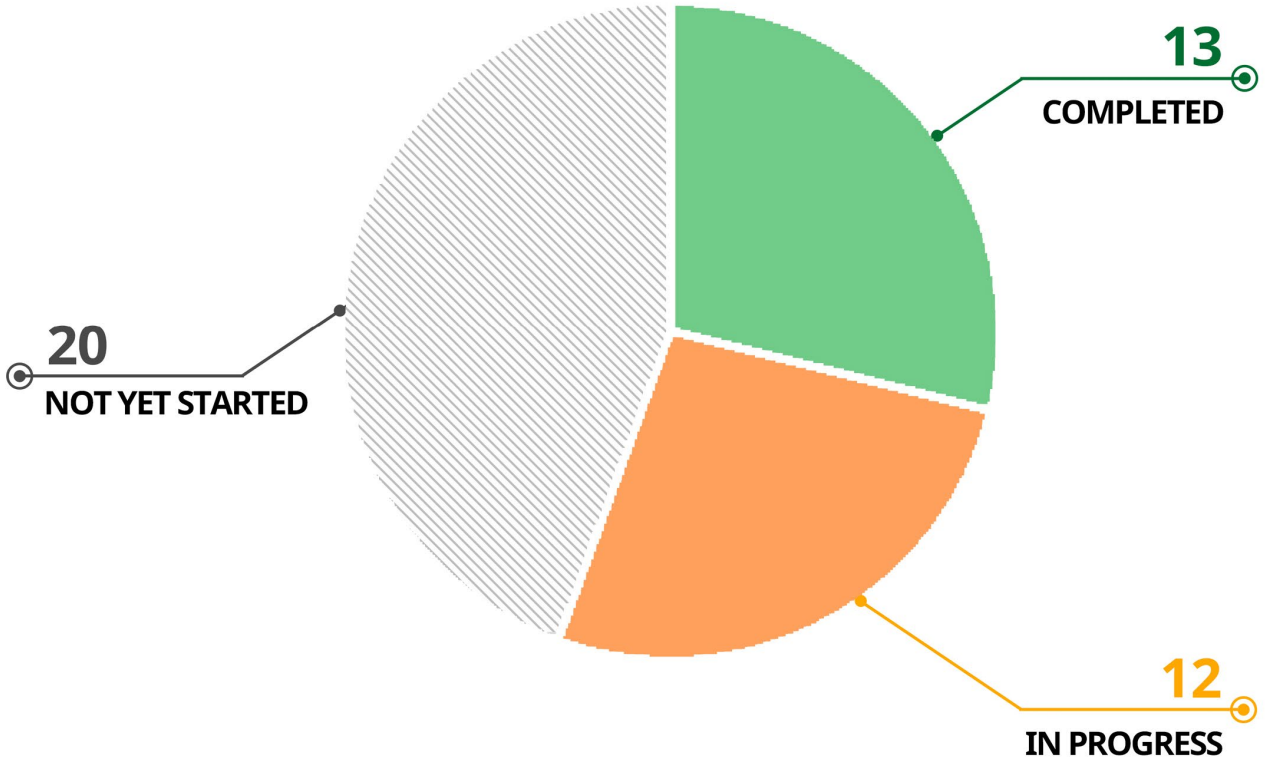


Low canopy cover (projected)

Measuring Progress — Long-term Plan

Of the recommended actions:

29% are completed
27% are in progress
44% are not yet started



Policy and Partnerships: Parking facilities

- 6.48.1 On grade, open parking facilities which contain five or more parking spaces must have at least five percent of the interior area of the parking facility landscaped
  - Each planting area to be at least 25’ long and contain at least one tree.
  - At least one tree for every ten parking spaces.

- Green Factor only comes into play with new surface lots.

- Parking lot ordinance applies to newly striped parking lots and new parking lots.

Policy and Partnerships: Parking facilities

- Modify the parking lot ordinance to include more specific canopy tree planting requirements (percentage of canopy cover at maturity and minimum soil requirements).

- Provide funding (tax rebates, grants, proceeds from the Tree Fund) for planting in existing parking lots.

- Create a database of parking lot owners amenable to accepting more planting on their parking lots that can provide off-site in-kind mitigation for TPO removals instead of mitigation payments to the Tree Fund.